The UAP Assessment Matrix: A proposed framework for evaluating evidence and understanding regarding Unidentified Anomalous Phenomena

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Abstract

Over recent years the issue of Unidentified Anomalous Phenomena (UAP) has increasingly captivated attention and even concern, as reflected in the US military establishing a UAP Task Force in 2020. By their very definition however, such phenomena present an epistemological challenge to observers and analysts, raising questions such as what does it *mean* for something to be unidentified or anomalous, and relatedly, what kind of evidence and understanding would it take for the phenomenon to become identified and explained. This paper aims to help address these issues by providing a UAP Assessment Matrix that would allow observers to appraise a given UAP event/case, featuring two main dimensions: evidence (i.e., the quality of the data pertaining to it); and understanding (i.e., the extent to which the data aligns with various theories and explanations). Moreover, both dimensions feature numerous sub-dimensions (which is what makes the framework a *matrix*), allowing more nuanced and fine-grained assessments to be made. We also demonstrate the matrix using a little-known but significant UAP case study from 1953. The matrix will ideally provide a foundation for more rigorous and considered analyses of UAP events and stimulate further understanding of this vitally important topic.

Key words: UAP; unidentified; anomalous; epistemology; assessment.

The UAP Question

For centuries, people worldwide have observed aerial phenomena that seemed "anomalous," some of which crucially we today – with our more advanced technologies and scientific understanding – might still regard as extraordinary (Vallée, 2008; Lomas & Case, 2023). During the 20th Century these attracted the label Unidentified Flying Object (UFO), and more recently UAP, with the latter now expanding even further to denote unidentified *anomalous* phenomena (given that some UAP appear to have transmedium capabilities, including traveling underwater). However, although there have been many sightings over the years – with the Mutual UFO Network receiving over 200,000 reports since 1969 (Mellon, 2022) – such observations have generally not been taken seriously by authorities, at least in public. (Whether authorities have privately been concerned is another matter, with indications that perhaps they have, as alluded to by Rear Admiral Hillenkoetter – first Director of the CIA, from 1947 to 1950 – whom the New York Times (1960) quote as saying, "Behind the scenes, high-ranking Air Force officers are soberly concerned about the UFO's. But through official secrecy and ridicule, many citizens are led to believe the unknown flying objects are nonsense.") This public disavowal began to change in 2017 though, when footage of three US military encounters with UAP was published online, bringing the topic to wider attention, especially following an article by Cooper et al. (2017) in the *New York Times*. The military angle was especially significant, since it involves observers who excel in occupations that require the highest skill and training in visual perception and processing, and whose testimony moreover is often triangulated with other information sources (e.g., radar). As a result, with the Department of Defense (2020) confirming the footage was genuine, soon after it established a UAP Task Force to investigate such incidents, a body which since July 2022 has been known as the All-Domain Anomaly Resolution Office (AARO).

AARO has since produced four reports (Office of the Director of National Intelligence, 2021, 2023a, 2023b; 2024), albeit with limited scope (mainly events since 2019 and focusing on US airspace and waters). In its latest report in November 2024, AARO stated that it has studied 1652 reports in total, with the report itself covering 757 (485 occurring during the latest reporting period of May 1, 2023 to June 1, 2024, plus 272 not included in previous annual reports). Of these 757, 49 cases were resolved as prosaic objects (e.g., balloons), with a further 243 cases recommended for closure (pending peer review), and 444 cases lacking sufficient data to facilitate analysis and placed in an Active Archive. Most intriguingly though, 21 cases were deemed to merit further analysis by AARO's intelligence community and science and technology partners. Strikingly, these include three reports

involving “pilots being trailed or shadowed by UAP,” of which notably “AARO has no indication or confirmation that these activities are attributable to foreign adversaries.” Speaking in May 2023, Dr Sean Kirkpatrick, AARO’s Director at the time (who left the post in December 2023), said he suspected most events *did* have conventional explanations, and only remained unidentified through lack of good data. However, around 2-5% were “possibly really anomalous” (Wendling, 2023). Similarly, in a briefing at the Pentagon on November 14th, 2024, to accompany the latest report, AARO’s new director, Dr Jon Kosloski, said, “we’re focusing on the truly anomalous where we don’t understand the activity,” noting that “there are interesting cases that I — with my physics and engineering background and time in the [intelligence community] — I do not understand and I don’t know anybody else who understands” (Vincent, 2024). Providing a little more detail/context, in an interview with (PBS NewsHour, 2025) in May 2025, Kosloski said that of the “little over 1,800 cases” that AARO current has in its holdings, about 2% have “sufficient scientific evidence that we can conduct a thorough investigation and after that investigation still remain anomalous.” For such cases, he suggested they “send it off to our scientific and intelligence partners for a very thorough analysis that generally takes on the order of months, and sometimes we don't even close those cases, we have to sort of keep them on the shelf always looking for additional data to enrich them because the phenomena are so perplexing.” A video of one such case, for example, was publicly shared by (All-domain Anomaly Resolution Office, 2025) on X in May 2025, with the accompanying text: "Eight minutes and fifteen seconds of video footage was captured by an infrared sensor aboard a U.S. platform in the Middle East in 2023 and later reported to AARO as UAP. The report remains unresolved as the available data does not support a conclusive analytic evaluation.”

Moreover, there are suggestions that AARO may have, at least historically, been deliberately downplaying the extraordinary nature of some evidence, per comparable previous investigations into UAP (such as Project Blue Book). Boswell (2022) reports a source in the Office of the Director of National Intelligence saying, “They're patting themselves on the back that they've resolved over half of them… But we don't give a crap about the ones they've resolved. Yeah, there's balloons up there, and balloons are sometimes mistaken for UAP. But there are s***loads of classified videos that are pretty profound and pretty clear. They don't want to talk about this stuff, because they really, really don't know what the hell they are.” Indeed, this lack of knowledge and understanding has been publicly alluded to by key figures. John Ratcliffe, for example, former Director of National Intelligence and now Director of the CIA in the new Trump Administration, said in 2021 “we

are talking about objects that have been seen by Navy or Air Force pilots, or have been picked up by satellite imagery, that frankly engage in actions that are difficult to explain, movements that are hard to replicate, that we don't have the technology for" (Lewis-Kraus, 2021). Moreover, despite efforts to investigate such phenomena, some are still perplexing authorities. Speaking in August 2023, for instance, General Mark Milley, Chairman of the Joint Chiefs of Staff, said the military "can explain a lot of it, but there is some that's really kind of weird and unexplainable" (Wolfgang, 2023). Even more recently, in September 2024, former White House National Security Advisor, retired Lieutenant General H. R. McMaster, said "There are phenomena that have been witnessed by multiple people that are just inexplicable by any kind of science available to us" (McHardy, 2024). Similarly, in November 2024, Senator Kirsten Gillibrand, one of just six senators on both the Intelligence and Armed Services committees, said of at least some UAP: "We don't know whose they are. We don't know what propulsion they use. We don't know the tech … It's not off the shelf stuff… This is a body of stuff nobody knows what it is" (Laslo, 2024).

Moreover, this issue made international headlines in late 2024 as the US East Coast, especially New Jersey, witnessed a mysterious wave of "drone" sightings which, at the time of writing (January 2025) still appear to be ongoing and remain unresolved. Of particular concern, as reported by Von Rennenkampff (2025), is that these have "hovered with impunity above sensitive government facilities, spurring urgent briefings at the White House." These events follow similar incidents in recent years, perhaps most prominently at Langley Air Force Base in Virginia, where for a period of 17 nights in late 2023, between one and two dozen brightly-lit objects – some estimated to be around 20 feet long and flying at over 100 miles an hour – flew at any one time, "creating a sense of chaos" (Lubold, 2024). According to Senator Gillibrand, these objects involved a "type of technology that our radar did not detect them arriving," and stated that the government does not know "what technology [the objects] are using." Moreover, government documents indicate that all counter-drone technology deployed against the objects "failed." Having observed the incursions personally, Gen. Mark Kelly, a decorated Air Force pilot and the commander of Air Combat Command, likened the incursions to the famous UAP-related Spielberg film, describing them as "Close Encounters at Langley." It is of course possible that some of the "craft" witnessed at Langley and over the East Coast more recently genuinely are "drones" (in the sense of human-made unmanned aerial vehicles), and indeed it is likely that geopolitical adversaries may well be using such technology to spy on US military assets. But observers of these events seem skeptical that such drones do account for *all* the observed activity, especially since some

objects seem distinctly different in appearance and capabilities from known human-made drones, including lacking wings or visible means of propulsion, sometimes being very large in size (e.g., as big as an SUV), and staying aloft for hours. Moreover, some people have made a connection between the drones and strange "orbs" of light, which for decades have been a hallmark of the UAP topic: Michael Melham, the mayor of Belleville, New Jersey, said he had seen footage of "glowing orbs of light, on camera, that as you watch them, right before your eyes, they seem to transform into a mechanical drone" (News Nation, 2025). As Federal Aviation Administration Administrator Steve Dickson for example noted, "Not too long ago we would have called these 'UFO's,'" while his chief of staff responded, "Yep! Now everything is a drone!"

Amidst the upsurge of public interest and official openness in the topic there is now an emergent academic literature investigating various aspects of the phenomenon. This itself is a relatively new development. From 1947 onwards the US and other countries experienced numerous waves of UAP sightings, leading to keen public and academic curiosity. However, after repeated attempts by authorities to downplay the issue – as reflected in the Hillenkoeter quote above – an authoritative report commissioned by the US Air Force and led by physicist Edward Condon (1969) effectively closed down the issue. Despite a substantial minority of sightings in the report resisting prosaic explanation, its conclusion was "that nothing has come from the study of UFOs in the past 21 years that has added to scientific knowledge." Soon after, key scientific institutions endorsed this dismissal, as reflected in a scornful editorial in *Nature* (1969) which said it would "be inapt to compare it with earlier centuries' attempts to calculate how many angels could balance on the point of a pin; it is more like taking a sledgehammer to crack a nut, except that the nuts will be quite immune to its impact." The consequence, as astronomer J. Allen Hynek (1977) put it, was the "coup de grace to the UFO era. Science had spoken. UFOs didn't exist, and the thousands of people who had reported strange sightings… could all be discounted as deluded, hoaxers, or mentally unbalanced" (p.287). From then on, the study of UAP effectively became taboo within mainstream academia (Wendt & Duvall, 2008), with "ufology" derided as a "pseudoscience" (Eghigian, 2017), and UAP themselves relegated to the off-limits epistemological category of the paranormal (Lomas, 2024). To an extent, this attitude still persists in some quarters, as reflected in an article in Scientific American by Kloor (2023), titled "How wealthy UFO fans helped fuel fringe beliefs: There is a long U.S. legacy of plutocrat-funded pseudoscience. Congress just embraced it."

In light of recent developments though, the tide may now be gradually shifting, with Scientific American more recently publishing a piece by Eghigian and Peters (2023) titled, "It's time to hear from social scientists about UFOs," which argued, "We don't conclusively know if UAP physically exist beyond the mundane, but we do know this: UFOs are social facts… Social scientists should weigh in on UAP, now." One indication of increased academic openness to the topic is provided by Yingling et al. (2023) who surveyed 1460 faculty at elite US institutions in early 2022. Although most were not particularly aware of the issue (when asked "in the past several years how often have you heard news about the UAP topic?," 85% said either never [6.3%], rarely [30.3%] or just occasionally [48.7%]), in terms of *curiosity* about it, only 17.2% were "not curious," with 25.4 % "slightly," 25.3% "moderately," 16.8% "very," and 15.3% "extremely" curious. There may still be institutional pressures that ward academics off studying the topic. A follow up analysis by Yingling and Yingling (2024) revealed that 52.7% reported some degree of concern for tenure or promotion if pursuing the subject academically, while 69% were concerned about potential ridicule. Despite such concerns though, this budding curiosity among scholars is beginning to manifest in a new wave of UAP-related papers in credible journals. These include, for instance, an environmental analysis of public UAP sightings and sky view potential (Medina et al., 2023), a survey on psychological aspects characterizing UAP witnesses (De la Torre, 2024), an analysis of the cross-cultural and global nature of UAP reports (Lomas, 2023), and a qualitative exploration of the potential for UAP "disclosure" to precipitate "ontological shock" among the public (Lomas, 2025).

The present paper seeks to add to this emergent literature by offering an evidentiary heuristic that allows consideration of possible explanations for UAP, and indeed "weird and unexplainable" events more generally. Anyone who has observed the UAP "scene" and discourse for any length of time will be accustomed to the interest and indeed passion the topic generates among some people. However, these discussions tend to unfold without a guiding rubric or map that can help evaluate the evidence at hand. There is thus a risk of people either sticking to entrenched positions and preconceptions, or alternatively – for those who are *not* entrenched – simply being unsure how to approach and weigh up the data to reach a viable conclusion.

To elaborate, at the risk of overgeneralizing, the community of people interested in the UAP topic can generally be grouped into several categories, which can roughly be arranged on a kind of spectrum featuring involving oppositional forms of certainty on either pole, with gradations of uncertainty and doubt in between. On the "left-hand" pole (say), are

those who are essentially certain that all UAP have a prosaic explanation. Having become convinced of various relevant assumptions (e.g., that a UAP of extraterrestrial origin is extremely unlikely due to the difficulties of interstellar travel), this group has all-but decided that an extraordinary explanation is near-impossible. We say "all-but," since even the most dogmatic person here would surely revise their stance if the evidence was compelling enough by whatever stringent standards they might set for themselves (such as the trope of a UAP landing "on the White House lawn"). Among this group one might find "debunkers," a subset of the certain who are not only confident in their assessment but are motivated to dismantle, downplay, or deny any evidence that may be suggestive of an extraordinary explanation. Still on the skeptical left-hand side of the spectrum are those somewhere between certain rejection and the mid-point of uncertainty-doubt. Often referred to as "skeptics," these are theoretically open to UAP having an extraordinary explanation, but have a very high bar in terms of evidence needed to accept it. Hovering around the mid-point are those who fundamentally cannot decide, or don't have a position/theory, or perhaps just don't even care. Moving into the positive territory, between the mid-point of uncertainty and the right-hand pole of positive certainty are those who lean towards some percentage of UAP having an extraordinary explanation, but still entertain substantive reservations and would require further compelling evidence to assert their position. Finally, at the positive right-hand pole are those who are totally certain that some UAP are extraordinary, perhaps because of some personal encounter or experience (with such people often referred to as "experiencers").

We suggest all these groups would benefit from an evidentiary heuristic to guide them. For those at the poles of certainty, such a framework may give them pause in their conviction, and encourage a more open-minded and cautious (in either direction) approach to the issue. Then, for those people characterised by various degrees of uncertainty, a framework could help them better approach the topic with an empirical rigour that may be helpful. To this end, we have created a "UAP Assessment Matrix" (UAP-AM) to help people engage with this topic. It involves two main axes: the vertical axis concerns the quality of evidence regarding a particular UAP event; and the horizontal axis pertains to the extent to which this data can be understood using various explanations and theories. Alternative labels for these axes could be also used though, including: truth (vertical) and belief (horizontal); data (vertical) and theory (horizontal); the phenomenon (vertical) and our reaction (horizontal); ontology (vertical) and epistemology (horizontal); and "external" (i.e., what is out in the world) (vertical) and "internal" (i.e., what is in people's minds) (horizontal). We should emphasize that the matrix is intended to articulate a given person's perspective on a

particular UAP event. There would therefore be a different iteration of the matrix not only for every event, but for every person's perspective on that one event. To illustrate this point, below we take a specific UAP case study, and each author offers their particular version of the matrix in response. Also though, we show it is possible to combine the matrices of multiple people (e.g., by averaging their scores) to create an overall consensus matrix. In the Conclusion we offer some practical suggestions for how this could work (e.g., as a useable tool on a publicly-available website). Before we introduce our matrix itself though, we provide some additional context by first exploring considering the "informational asymmetry" involved in the UAP topic.

Information Asymmetry and Complexity

Before delving into the UAP-AM, we must note that this issue involves a highly complex informational landscape, and everyone approaches it from a different epistemological perspective. This is perhaps true of any topic: whatever the phenomenon under consideration, no two people will have identical knowledge, information, beliefs, values, assumptions, memories, and so on. However, this epistemological splintering is acutely exacerbated with this arena in particular, which is arguably unparalleled in the extent to which it involves a complex, contested, and above all *asymmetric* information landscape. Essentially, so much about this area is not public knowledge, with most of the information possessed by authorities being highly classified. Going further, it is highly compartmentalised and restricted even to figures in authority with the highest security clearances, and may well never be fully known by any one person. Despite these epistemological walls though, some information has leaked into the public domain in some way, such as via people in authority divulging apparent details or insights. However, this story is very slippery and uncertain, full of rumours and allegations that are often never fully substantiated. In all its aspects, people are thus required to make judgement calls about the veracity of claims, while never being fully certain of the true picture.

For example, June 2023 saw public allegations by David Grusch, an intelligence community whistleblower, that US authorities have maintained a secret UAP "crash retrieval" and "reverse engineering" program dating back to a UAP crash in Italy in 1933 (Kean & Blumenthal, 2023). There is no way of knowing the truth of his allegations at the time of writing. Significantly though, Senator Marco Rubio, previously Vice Chairman of the Senate Select Committee on Intelligence and now Secretary of State in the new Trump Administration, said in an interview on June 26 2023 that "many" other whistleblowers had testified corroborating Grusch's claims, including people with "firsthand knowledge" of such

programs, most of whom hold (or held) "very high clearances and high positions within our government." Corroborating this point, Shellenberger et al. (2023) similarly claimed "at least 30 other whistleblowers" had given such testimony to various authorities. Some significant figures who would seemingly be well placed to judge Grusch's assertions have even vouched for them publicly. Retired Army Colonel Karl Nell for example – who served alongside Grusch in the UAP Task Force – said "His assertion concerning the existence of a terrestrial arms race occurring sub-rosa over the past 80 years focused on reverse engineering technologies of unknown origin is fundamentally correct" (Von Rennenkampff, 2023b). Other significant figures have also seemingly corroborated some of Grusch's claims (without necessarily vouching for Grusch himself). Dr. Harald Malmgren, for instance, who served as an advisor to Presidents Kennedy, Johnson, Nixon and Ford, said in 2024 that Richard Bissell (CIA Deputy Director for Plans, 1959-1962), had briefed him on "otherworld technologies," and elaborated in a recent interview that these included the 1933 Italian crash, which was purportedly transferred into the possession of the US in the mid-1940s (Michels, 2025). Equally strikingly, Malmgren said that he personally handled material from a UAP apparently downed in relation to a high-altitude nuclear test conducted over the Pacific in October 1962 (Cruikshank, 2024).

Furthermore, Grusch's allegations appear to have prompted or influenced legislative efforts around these issues, most notably bipartisan proposals, introduced in July 2023 by Chuck Schumer, Majority Leader of the Senate, for an Unidentified Anomalous Phenomena Disclosure Act (U.S.C. S.2226, 2023). As reported in The New York Times, this would create "a commission with broad authority to declassify government documents about U.F.O.s and extraterrestrial matters, in an attempt to force the government to share all that it knows about unidentified phenomena" (Barnes, 2023). Strikingly, the legislation essentially reiterated Grusch's claims, including defining "legacy program" (i.e., efforts to "collect, exploit, or reverse engineer technologies of unknown origin"). In the event, the provisions in the Act were significantly weakened when only minimal aspects were passed into law in December within the National Defense Authorization Act (NDAA) for the fiscal year 2024, of which it was a proposed amendment. Notably, resistance reportedly came from House Representatives with close ties to some of the companies rumoured to have helped engage in UAP reverse engineering efforts, and hence who would be affected by, and moreover motivated to oppose, the original legislation. As Schumer (2023) himself posted on X, "It is an outrage that the House didn't work with us on our UAP proposal for a review board [one of the most consequential aspects of the proposed legislation]. This means declassification of

UAP records will be up to the same entities who have blocked and obfuscated their disclosure for decades. We will keep working to change the status quo." Moreover, in a colloquy on the Senate Floor with Senator Rounds on December 13, Schumer said, "We have also been notified by multiple credible sources that information on UAPs has also been withheld from Congress, which if true is a violation of laws requiring full notification to the legislative branch – especially as it relates to the four congressional leaders, the defense committees, and the intelligence committee." Crucially, it seems highly unlikely the Senate Majority Leader would make such public remarks, and especially propose such remarkable legislation, absent highly credible corroborating evidence or testimony.

Even more relevantly though, that a politician as powerful as the Senate Majority Leader appears unable to access or disseminate records on this topic illustrates the point about information complexity and asymmetry we are making here. So too does the fact that we cannot assess the veracity of Grusch's allegations, so need to regard his claims as being *possibly* true, while recognising they may well not be. Hence the point of raising the Grusch situation here, since these considerations apply to the topic as a whole, for which he serves as a timely and vivid case study, with the whole area shrouded in a bewildering epistemological murkiness. Accordingly, the discourse in this arena is replete with terms that allude to this. These include "cover-up" (i.e., allegations that authorities are withholding information from the public), and relatedly "disclosure" (i.e., the notion that authorities will admit and reveal this information), with the latter even featuring in the very title of Schumer's proposed amendment. There are also complex concepts like "conspiracy," which can be ambiguous, applying both to allegations of cover-ups (i.e., implying these allegations are a false "conspiracy theory"), and to potential cover-ups themselves (i.e., implying that people in authority have conspired to do the cover-up), as per Grusch's claim. The picture becomes more complex still when appreciating that "authorities" is not a monolith, but a fragmented system with competing agendas, as reflected in the fact that Schumer – arguably, after the President, the most powerful politician in the country at the time – was attempting to initiate legislation that would compel his own party (among others) to release UAP information to the public. As Schumer wrote in explaining his motives for introducing the Act, "The American public has a right to learn about technologies of unknown origins, non-human intelligence, and unexplainable phenomena" (Bolton, 2023). As this quote illustrates, this is a complex information landscape, involving myriad parties with different epistemological positions and perspectives, creating an uneven playing field which has been referred in other contexts as information asymmetry (Mishra et al., 1998).

All these considerations have relevance to the matrix, in various ways. As we explore shortly, one axis involves a person's appraisal of the quality of evidence for a given UAP case. First, people differ in their *ability* to appraise the evidence, for myriad reasons, some of which are connected to these concerns around information asymmetry. One difference is access to information. People with security clearances to access classified information will have a more complete set of information on any given event, and moreover this data may be more detailed (e.g., full-length high-definition videos, in contrast to clipped, lower-definition videos that may be available to the general public). This important point is often overlooked by people who assume that just because they don't have access to the data then it doesn't exist. Seth Shostak (2023), for example, senior astronomer at the SETI Institute, wrote an article for MSNBC essentially mocking Grusch's claims. Asking, "Where is the evidence?", he concluded, "It's MIA. Neither Grusch nor anyone else claiming to have knowledge of secret government UAP programs has ever been able to publicly produce convincing photos showing alien hardware… The believers maintain that such photos exist but are being kept under wraps. For reasons that are always unclear, the critical evidence that would convince anyone… can't be made public." Given the manifold security implications of divulging such evidence, it seems reasonable and logical that it *would* be classified, which is what Grusch has alleged. In any case, the pertinent point here is that people have differential access to information, with implications for their assessment of the situation. Another difference is training and expertise; some people will be more skilled at assessing the data based on intellectual or professional competence (e.g., in reviewing video evidence). This means not all attempts at the matrix will be equal: some people's takes will more accurate than others, so should be assigned greater weight in some way, as we discuss in the Conclusion.

A yet more complicated issue relating to information asymmetry concerns the *quality* of the evidence. There are various considerations relating to quality, as we explore below, but perhaps the most contentious here is its *reliability*. The discourse around this topic is steeped with accusations relating to charges such as misinformation and disinformation. The Grusch situation is a case in point. Even while he has testified to his claims under oath, some critics still suggest he could be offering false or misleading information, for example as part of an elaborate "psy-op." This could either be done knowingly on his part (i.e., he is an active agent in the psy-op), or unknowingly (i.e., he believes the information to be true but is himself the victim of a psy-op). Of course, these scenarios would be extraordinary in their own ways, as articulated by Von Rennenkampff (2023a), in an article titled, "A monumental UFO scandal is looming", which quoted then-Senator Marco Rubio as saying, "Either what [Grusch] is

saying is partially true or entirely true, or we have some really smart, educated people with high clearances and very important positions in our government who are crazy and are leading us on a goose chase." As Von Rennenkampff puts it, "Either the U.S. government has mounted an extraordinary, decades-long coverup of UFO retrieval and reverse-engineering activities, or elements of the defense and intelligence establishment are engaging in a staggeringly brazen psychological disinformation campaign," both of which would be highly problematic, hence the article title. The main point here is that information on this topic cannot be simply accepted at face value, but must always be viewed through a lens of skepticism in terms of whether (or to what degree) it is true, and what agendas and motives the people divulging the information may have. So, while whistleblower claims can certainly be datapoints that might be factored into the matrix, one must do so carefully and judiciously. These considerations are reflected in the way we have designed its rating system, for example placing a higher value on "rumours" heard from sources that one deems "reliable" versus "unreliable."

The Matrix

Having laid out the context and reason for our UAP-Assessment Matrix (UAP-AM), we now introduce the matrix itself. The first point to note is that, being a framework for appraising evidence relating to UAP, it is designed to accommodate different forms of evidence. To that point, as we discuss in a detailed reflection in Appendix 1, across myriad fields of human endeavour concerned with seeking truth and evaluating claims in that regard, evidence is widely considered most valuable if it is verifiable, reproducible, and objectively analyzed, thereby contributing meaningfully to the hypothesis testing process. We must recognize though that evidence comes in various forms, and relevant fields have developed protocols to weigh and adjudicate types of data in their particular domain. To give a salient example, in many areas of science, a phenomenon experienced by just a single person may well be dismissed as unreproducible anecdote, and not given much standing. But in a court of law, by contrast, eye-witness testimony, even of a single person (if credible), carries considerable weight. Relatedly, the legal system has developed useful principles such as acquittal on the basis of "reasonable doubt." Legal theory and practice have also drawn extensively on research into the fallibility of human witnesses in terms of accurate perception, memory, and communication of information, as seem perhaps above all in the work of Elizabeth Loftus (2018). Our matrix is intended to allow all these considerations to be brought to bear on the evaluation process.

Indeed, while we ourselves are scientists, we suggest that perhaps the best way to approach this topic and engage with the matrix is through a *criminal justice* lens, as if a detective solving a crime. While there are of course intellectual and practical overlaps between scientific and criminalistic endeavours, there are important differences, and engaging with the UAP topic seems closer in spirit to the latter. This point was made by renowned UAP scholar Hal Puthoff (2022), who argued that "it would behoove us to set aside our tentativeness about any given hypothesis, place it on the table as almost certainly true, and "go for it"", doing so "by incorporating elements of a more forensic "gumshoe" style, as in criminalistics and intelligence work." Conventional scientific experimentation involves hypothesis-testing through controlled manipulation of variables and careful observation of the results. In the legal sphere, by contrast, the key event *has already happened*, and the task is to arrive at an explanation that best fits the surviving evidence. Let's call a UAP event itself the "crime" and the various potential hypotheses the "suspects." The task then is to ascertain which suspect seems the most reasonable perpetrator "beyond a reasonable doubt." For a start, this criminalistic mindset means one would necessarily adopt the kind of skeptical stance discussed above in relation to information asymmetry, whereby one would not just critically appraise the data but also the people providing it, just as a detective wouldn't take eye-witness testimony at face value but would interrogate it with due caution. Moreover, this mindset is perhaps more pragmatic and open to extraordinary possibilities than conventional science, which on the whole tends to be quite conservative in its unwillingness to challenge established theories (Kuhn, 1962). This open-minded pragmatism is reflected in the works of Sir Arthur Conan Doyle, which have numerous variations on the expression "When you have eliminated the impossible whatever remains, however improbable, must be the truth" (in The Sign of the Four, 1980). Essentially, follow the evidence without prejudice, and if that leads to an extraordinary conclusion, then so be it.

A second point to mention is that we mean "UAP" in the widest sense. More narrowly, this could be applied to UAP that fits the classical notion of a "UFO" (e.g., an unconventional craft in the sky), but could also apply to any phenomena that seemed anomalous and hard to identify in some way (e.g., strange weather phenomena). Here we introduce the matrix over three subsections. First, we explain its two main dimensions, then we present the UAP-AM itself.

The Two Dimensions

The foundation of the matrix is a two-dimensional state space involving a vertical and horizontal axis. What makes our model a *matrix* is that both axes in themselves have various

sub-dimensions, as opposed to each being a singular spectrum. Before introducing these sub-dimensions, we begin with the two main axes: (a) evidence (vertical) and (b) understanding (horizontal). As noted above though, other potential labels would be equally relevant and applicable, including: (a) truth and (b) belief; (a) data and (b) theory; (a) the phenomenon and (b) our reaction; (a) ontology (the nature of the phenomenon) and (b) epistemology (our understanding of it); and (a) "external" (i.e., what is out in the world) and (b) "internal" (i.e., what is in people's minds). The vertical dimension is the UAP event itself: whatever data and evidence we have about the phenomenon that occurred. The horizontal dimension is then our understanding and interpretation of that data in light of existing ideas, theories, expectations, and so on.

Of particular relevance in understanding these dimensions are the poles that define them. With the vertical "evidence" dimension, its poles pertain to the *quality* of the evidence, spanning an absence of evidence (at the very base of the dimension), to minimal quality (very low scores) to the highest possible quality (at the top). Quality itself is hard to define of course, and involves numerous considerations. This is precisely the point of the matrix, since any given judgement of quality can involve myriad elements. For now though, let us just imagine the epitome of low quality evidence being something like a UAP sighting that was witnessed by a single person, and this person was moreover drunk, had poor eyesight, and a reputation for dishonesty, and moreover the night of the observation was hazy and featured a meteor shower that could easily account for the apparent UAP. By contrast, a zenith of quality might be found in a UAP event involving not only hundreds of direct witnesses, but with these all of exemplary perceptual skills and professional responsibility (e.g., the crew of a battleship), and moreover their observations are triangulated with, and corroborated by, many other data points, such as radar.

Then, when it comes to the horizontal dimension, this is defined in reference to both (a) the evidence, and (b) our current level of understanding. As with the evidence, assessing such understanding is complex and multifaceted, hence the value of our matrix. But for now, let us grant it is possible to approximate a general consensus regarding a given aspect of scientific knowledge and technical expertise, and hence to appreciate if a given UAP event seems to contravene or challenge these prevailing standards. To that point, Lue Elizondo (2024), a former senior US intelligence official who previously helped run a Pentagon program tasked with investigating UAP, has suggested that "genuine" (i.e., truly anomalous UAP) are often characterized by one or more of "five observables": (1) sudden and instantaneous acceleration; (2) hypersonic speed without signatures (e.g., sonic booms); (3)

"anti-gravity" capacities (seemingly able to generate lift without any visible means of propulsion); (4) low observability (e.g., concealing or cloaking their presence); and (5) trans-medium travel (i.e., moving easily in and between different environments, such as space, earth's atmosphere, and even water). A slight complication here is that there are rumours that secretive advanced human-made craft may be capable of some of these capacities (Cook, 2023), hence potentially being responsible for some apparent UAP observations. That said, further complicating such allegations are claims that these very technological developments may be due, at least in part, to humans reverse engineering genuine (i.e., NHI) craft (Boswell, 2025), which somewhat blurs the line between whether any such possible cases should be attributed to human versus non-human feats. Nevertheless, such rumours aside, the general consensus still appears to be that our technologies cannot yet achieve these feats. As such, if any UAP were reliably assessed as exhibiting one or more of these observables one might be led towards the conclusion that it genuinely is "extraordinary" (i.e., beyond current human understanding and capabilities).

Consider for example the first observable. Even aside from humans currently lacking the technical capabilities to build craft with relatively instantaneous acceleration, one would also need to factor in scientific understanding around considerations such as physics and physiology, especially in relation to the phenomenon of "g force" (a measure of acceleration, where one g is equal to the acceleration caused by Earth's gravity, i.e., the acceleration of falling objects on Earth). In that regard, in terms of physiology, it is generally agreed that humans can usually only withstand up to 9 g's (although on very rare occasions people have survived extremely brief g forces into the 100s), and even then only for a few seconds, with even 6 g's potentially being fatal. This kind of knowledge provides the foundation for the horizontal dimension, the poles for which represent the extent to which the evidence conforms to this baseline of knowledge: from complete conformity on the left-hand side to complete non-conformity on the right-hand side. Staying with g force as an example, a UAP might be analysed as making manoeuvres that are calculated to produce particular levels of g. At the left-most edge of the spectrum, this might be a completely conventional level of 2-3 g's. A g force of 9 might place the observation at the middle of the spectrum: nearing the point where, if this is a craft with a pilot, we are swiftly approaching not being able to understand our observations in the context of current knowledge. Then, any level of g above 9 would take us into the right-hand territory, in which we do not understand how it would be possible for a pilot – if there was one – to survive such manoeuvres. Juxtaposing these two

spectra produces a two-dimensional state space with four quadrants, as visualized in Figure 1 below:

- High quality evidence in conformity with current understanding (upper left). This is the quotidian, conventional occurrences of normal life, such as an aeroplane passing overhead: a phenomenon with myriad data sources that completely aligns with current knowledge.
- Low quality evidence in conformity with current understanding (lower left). As per the upper right quadrant, this too involves conventional phenomena, except the evidence is of lower quality. For example, in relation to the present topic, in February 2023 emerged accounts of a Chinese spy balloon flying unauthorised over US airspace. Initially the evidence in support of this claim was low, but the phenomenon itself did not conflict with scientific understanding. This event would thus be located here.
- High quality evidence not in conformity with current understanding (upper right). Here we get to the heart of the UAP issue: aerial phenomena involving high quality evidence (e.g., multiple reliable witnesses backed up by other data sources) that seems extraordinary (i.e., not readily understood in light of conventional knowledge).
- Low quality evidence not in conformity with current understanding (lower right). Finally, in this quadrant are the inchoate clutter of anomalies that populate the wider discourse on the UAP topic: the myriad sightings and events of unusual phenomena that may be persuasive to those who are witnesses, but for which there is lacking quality evidence beyond their individual testimony. Given this lack of evidence, such reports are liable – unfairly or not – to be not taken seriously.

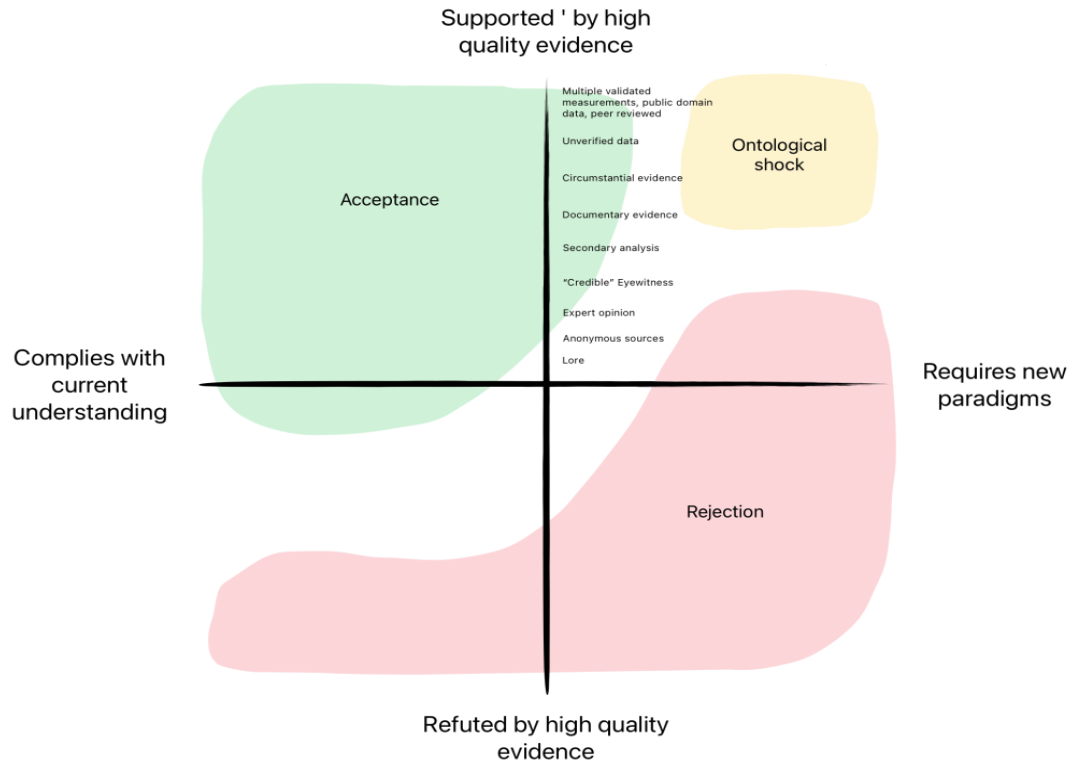


Figure 1. The Basic 2-Dimensional State Space Underpinning the UAP-AM.

We therefore have an initial 2-D state space that can help us appraise UAP observations. As highlighted above though, both the vertical and horizontal dimensions are themselves multi-faceted. As such, we can conduct a more nuanced appraisal by turning the state space into a multidimensional matrix, inspired by and drawing on a similar matrix developed in relation to human wellbeing (Lomas & VanderWeele, 2023).

The UAP-AM

Both evidence and understanding are not singular considerations, but each encompass myriad dimensions, as illustrated in Figure 2 below, which presents a "blank" (uncompleted version of the matrix). This section elucidates the dimensions of the matrix in the form of a questionnaire that could be given to people regarding a given UAP event. The paper then concludes by presenting a real historical event, which we then evaluate using the matrix, as detailed in Figure 3.

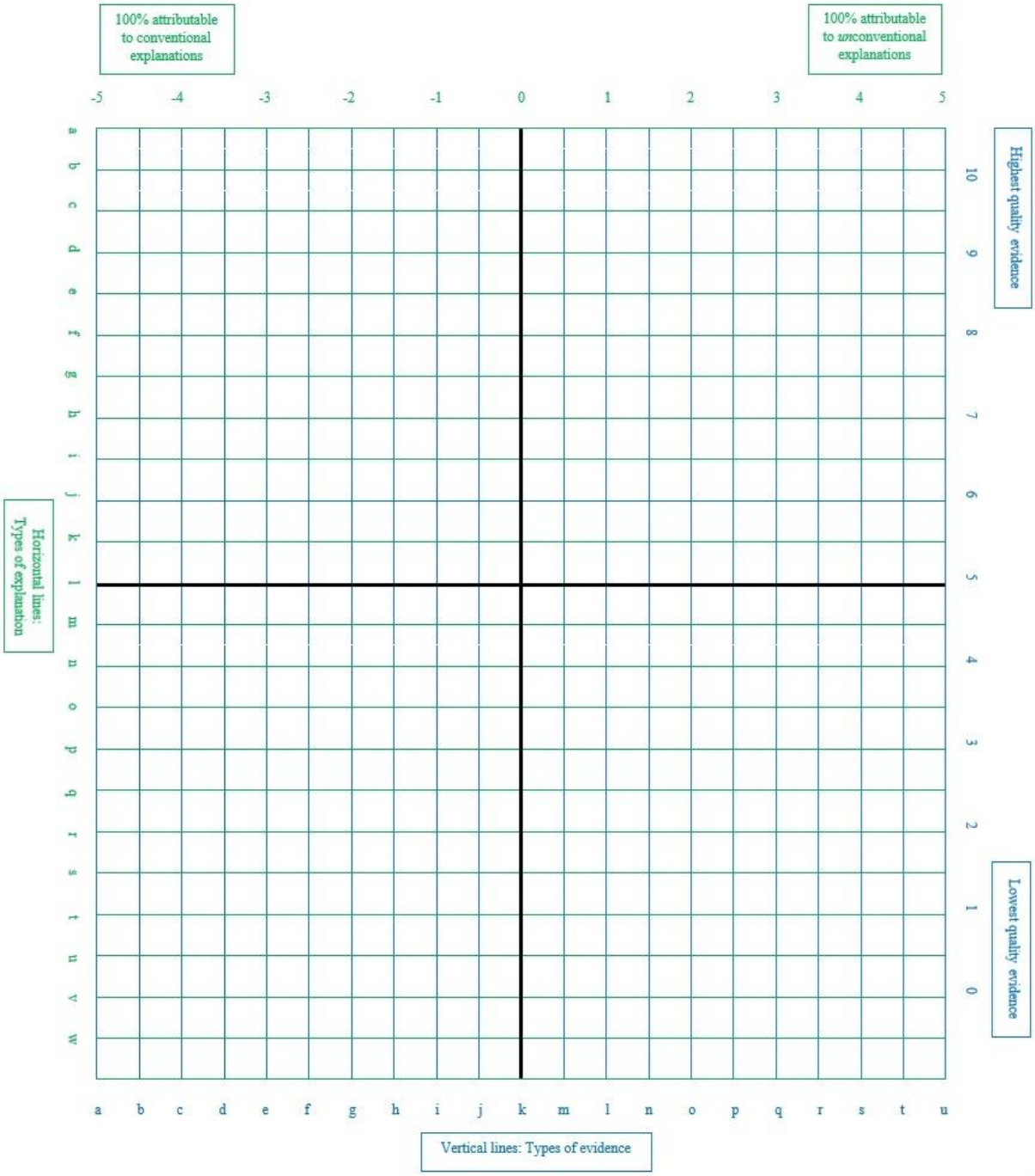


Figure 2. The UAP-AM

The UAP-AM questionnaire has two main sections. The first section pertains to the quality of the evidence, and constitutes the vertical dimensions of the matrix. The second section pertains to the extent to which the evidence can be accounted for by conventional explanations and scientific knowledge, and constitutes the horizontal dimension of the matrix. Let's begin with section 1, covering the quality of the evidence. Below is a list of components that collectively could be involved in appraising evidence pertaining to a given

UAP incident. These considerations can all be framed as a question that invites an answer on a 0-10 scale. The instructions for someone answering these questions are as follows:

> "First, we would like to ask your opinion about the existence and quality of the evidence pertaining to the UAP event in question. With respect to each type of evidence, please rate this in terms of value/merit on a scale from 0 (no value/merit whatsoever) to 10 (the highest possible value/merit). This judgement is subjective and relative, but please nonetheless attempt to give a specific number in each case. Note that a *confirmed absence* of a given form of data (e.g., if you are certain that no photographic evidence exists) automatically constitutes a 0. By contrast, if you *cannot answer* a particular question (e.g., because you do not personally *know* if a given form of data exists), please just write "N/A" ("not answerable"). Moreover, each dimension is separate, so the profile overall can reflect the nuances of the data. For example, the first dimension is the number of direct witnesses; having a thousand witnesses might mean this dimension is rated as a 10, whereas only having one witness might mean it is rated as a 1. However, in the latter case, this particular witness may be of the highest quality possible, and so the second dimension might be rated as a 10. Note that, for most questions, a guide is provided below it to help you answer.

- Va. Amount of direct witnesses
    - 0 = You know for certain there are no direct witnesses
    - 1-2 = You have heard rumours there are a few direct witnesses
    - 3-4 = You have heard rumours there are many (e.g., >100) direct witnesses
    - 5-6 = You know for certain there are a few direct witnesses
    - 7-8 = You know for certain there are many (e.g., >100) direct witnesses
    - 9-10 - You know for certain there are huge numbers of witnesses (e.g., >1,000)
- Vb. Quality of direct witnesses
    - If you answered 0 to Va, skip this question
    - 0 = The witnesses are the worst possible quality (e.g., known liars with no expertise in the UAP topic)
    - 10 = The witnesses are of the highest possible quality (e.g., witnesses of unimpeachable character with world-leading expertise in the UAP topic)
- Vc. Amount of secondary witnesses (e.g., saw radar footage or other data)
    - Same reference points as Va
- Vd. Quality of secondary witnesses

    - Same reference points as Vb
- Ve. Triangulation (i.e., “agreement”) among witnesses
    - 0: The witnesses directly contradict each other on all important details
    - 1-2: The witnesses directly contradict each other on many important details
    - 3-4: The witnesses directly contradict each other on a few important details
    - 5-6: The witnesses are broadly in agreement but have slight differences in many respects
    - 7-8: The witnesses are broadly in agreement but have slight differences in a few respects
    - 9-10: The witnesses are in complete agreement in all respects
- Vf. Existence of photographic data
    - 0 = You know for certain there is no photographic data
    - 1-2 = You have heard rumours from unreliable sources there is photographic data
    - 3-4 = You have heard rumours from reliable sources there is photographic data
    - 5-6 = It is widely agreed and well-established there is photographic data
    - 7-8 = You yourself have seen photographic data but you are unsure about its reliability
    - 9-10 = You yourself have seen photographic data that you trust 100%
- Vg. Quality of photographic data
    - If you answered 0 to Vf, skip this question
    - 0 = The worst possible quality photographic data (e.g., one shot, blurry, from a distance, etc.)
    - 10 = The highest possible quality photographic data (e.g., numerous shots, from many people/sources, at close range, and in perfect fidelity/clarity)
- Vh. Existence of video data
    - Same reference points as Vf
- Vi. Quality of video data
    - Same reference points as Vg
- Vj. Existence of radar data
    - Same reference points as Vf
- Vk. Quality of radar data
    - Same reference points as Vg
- Vl. Existence of infrared data (e.g., FLIR)

    - Same reference points as Vf
- Vm. Quality of infrared data
    - Same reference points as Vg
- Vn. Existence of other electromagnetic signals (i.e., other than infrared and visible light)
    - Same reference points as Vf
- Vo. Quality of other electromagnetic signals
    - Same reference points as Vg
- Vp. Existence of physical/material evidence (i.e., recovered material from UAP)
    - Same reference points as Vf
- Vq. Quality of physical/material evidence
    - Same reference points as Vg
- Vr. Existence of other data sources (i.e., other than those listed above)
    - Same reference points as Vf
- Vs. Quality of other data sources
    - Same reference points as Vg
- Vt. Triangulation (i.e., "agreement") among data sources
    - Same reference points as Ve
- Vu. Public availability of data
    - 0: No direct evidence is in the public domain
    - 10 = All direct evidence is in the public domain, is freely available, and is easily accessible

The scores for all these answers can each be plotted on the various vertical dimensions of the matrix, as will be illustrated below using a specific UAP case study. Also, we can compute their average score to give an overall indication of quality, with the higher the number, the greater the overall quality of the evidence. Additionally, in assessment efforts where time or space to ask numerous questions are limited, we can also condense these various considerations into a single-item question: According to your best knowledge and understanding of this case, on a scale of 0-10, please rate the quality of the evidence for it, where 0 represents the lowest possible quality evidence, and 10 the highest.

When it comes to the horizontal dimension of understanding, there are numerous different aspects, which can be ordered into three subsections: (1) conventional explanations; (2) unconventional explanations; and (3) current scientific knowledge. The first subsection,

represented by the first nine items below, concerns the plausibility of a phenomenon being accounted for by conventional (i.e., prosaic) hypotheses (e.g., human technology). The second, represented by the following six items below, is in reference to unconventional (i.e., extraordinary) hypotheses. Finally, the third subsection involves current scientific understanding and technology, focusing on nine fields. Again, all these considerations could be rated on a 10-point scale, except this time it ranges from -5 to + 5. Essentially, taking zero as a neutral starting point, the further into "negative" territory a person's responses go, the greater their confidence that the UAP event *can* be completely accounted for by conventional phenomena and completely explained by current science and technology (with -5 being total certainty). By contrast, the higher into "positive" territory their responses go, the greater their confidence that the event *cannot* be accounted for by conventional phenomena and current science and technology, and instead requires "extraordinary" explanations (with +5 being total certainty that the latter explanations are required). Then, bisecting these territories, zero represents a neutral middle ground, where one essentially is not sure or cannot commit either way.

Before introducing our items, we should emphasise that our categories of explanation are fairly broad, and it is possible to have a much finer-grained classification taxonomy. Consider a recent presentation by Retired Army Colonel Karl (Nell, 2023), who was mentioned above as serving alongside Grusch in the UAP Task Force and also publicly corroborating his allegations; he prepared a highly detailed schema involving *72* different hypotheses organised into three overarching categories and 14 sub-categories, as detailed in appendix 7. It is worth noting here that of the "extraordinary" hypotheses proposed for UAP by Nell and others, extraterrestrial explanations are not the only possibility. Another potential class of options are collectively known as "ultraterrestrial" hypotheses, which Puthoff defines as "sequestered terrestrial cultures… existing alongside us in distinct stealth" [38]. These include, for instance, highly speculative "extradimensional" explanations, in which rather than UAP and whatever beings may occupy them travelling to earth from another planet, they are conceived as potentially inhabiting and materialising from other (e.g., "higher") dimensions that coexist alongside the standard four of spacetime we humans usually perceive, hence ultra-terrestrial (i.e., such objects and beings are already present in Earth's environment in some sense, just not in ways we can conventionally understand). Expansively accounting for all these mooted possibilities, Nell's taxonomy is therefore highly detailed and granular; it includes, for example, four different varieties of the standard extraterrestrial hypothesis, whereas in our framework below these are condensed as a single hypothesis.

However, it is possible to make our matrix, and the accompanying assessment tool, as fine-grained as desired, and so it would be possible, for example, for it to include 72 hypotheses along the lines suggested by Nell. As it stands though, the version of the matrix here contains only 24 different dimensions pertaining to understanding. In that respect, beginning with the first section of the horizontal dimension (i.e., conventional explanations), the instructions for people answering the survey are as follows:

> "According to your best knowledge and understanding, on a scale of -5 - +5, please rate the likelihood that the case in question can be plausibly accounted for by the following conventional explanation. For reference: -5 = 100% confident that the case can be easily and completely accounted for by the particular explanation; 0 = not sure if the case can be accounted for by the particular explanation; +5 = 100% confident that the case is impossible to account for by the particular explanation."

- Ha. A physical psy-op (i.e., an incident genuinely occurred, but was staged by humans, intended to mislead observers into believing it is a "real," i.e., extraordinary, UAP)
- Hb. A hoax or lie (i.e., reports and evidence of the event are fabricated)
- Hc. Misperception (i.e., genuine physical stimuli were observed, but were misinterpreted)
- Hd. Hallucination (i.e., genuine physical stimuli were *not* observed, but were merely imagined)
- He. Conventional human aviation technology (e.g., a plane)
- Hf. Conventional airborne objects (i.e., something human-made that can be airborne but which we would not define as aviation technology, such as a "party balloon")
- Hg. Secret human aviation technology (e.g., a "black" military special access project)
- Hh. Explainable meteorological phenomena (i.e., a weather process)
- Hi. Explainable biological phenomena (i.e., an airborne organism such as a bird)

"Now we would like you to consider the likelihood that the case in question requires various *un*conventional explanations. In this case, please use the following such that: -5 = 100% confident that this hypothesis is completely untenable and invalid for the case in question; 0 = not sure if this hypothesis is tenable or valid for the case in question; +5 = 100% confident this hypothesis is unquestionably the correct one for the case in question."

- Hj. Extraterrestrial intelligences (i.e., entities from elsewhere in the cosmos)
- Hk. Extratempestrial intelligences (i.e., time travellers, such as future humans)

- Hl. Cryptoterrestrial intelligences (i.e., entities concealed within or near Earth)
- Hm. Interdimensional intelligences (i.e., inhabiting other dimensions beyond the conventional four of spacetime)
- Hn. Angelic intelligences (i.e., spiritual entities, as described in religious teachings)
- Ho. Unexplained meteorological phenomena (i.e., a weather process humans are unaware of)

"Finally, according to your best knowledge and understanding, please rate the likelihood that the case in question can be plausibly explained by current scientific theory/knowledge and technical expertise in the following fields. For reference: -5 = 100% confident the case in question can be easily and completely explained by current theory/knowledge and technical expertise in this particular field; 0 = not sure if the case in question can be explained by current scientific theory/knowledge and technical expertise in this particular field; +5 = 100% confident the case in question is impossible to explain by current scientific theory/knowledge and technical expertise in this particular field."

- Hp. Physics
- Hq. Material science and technology
- Hr. Aviation science and technology
- Hs. Biology
- Ht. Chemistry
- Hu. Psychology
- Hv. Anthropology
- Hw. Economics
- Hx. Politics

The scores for all these answers can also be plotted on the various horizontal dimensions of the matrix, as we illustrate below shortly. Also, we can compute their average score to give an overall indication of the most plausible explanation for the data, with the higher the number, the more the event would seem to be "extraordinary" (i.e., not attributable to conventional explanations, and/or challenging current scientific knowledge). Additionally, in assessment efforts where time or space to ask psychometric questions are limited, we can also condense these various considerations into a single-item question: According to your best

knowledge and understanding, on a scale of -5 - +5, please rate the likelihood that the case in question can be plausibly accounted for by conventional phenomena (e.g., weather, human technology, etc.) and current scientific understanding; For reference: -5 = 100% confident the case can be easily and completely accounted for by conventional phenomena and current scientific understanding; 0 = not sure if the case can be accounted for by conventional phenomena and current scientific understanding; +5 = 100% confident the case is impossible to account for by conventional phenomena and current scientific understanding. Now, to help illustrate the nature of the UAP-AM, the next section applies it to a particular UAP case.

Case Study: 1953 Lockheed Sightings

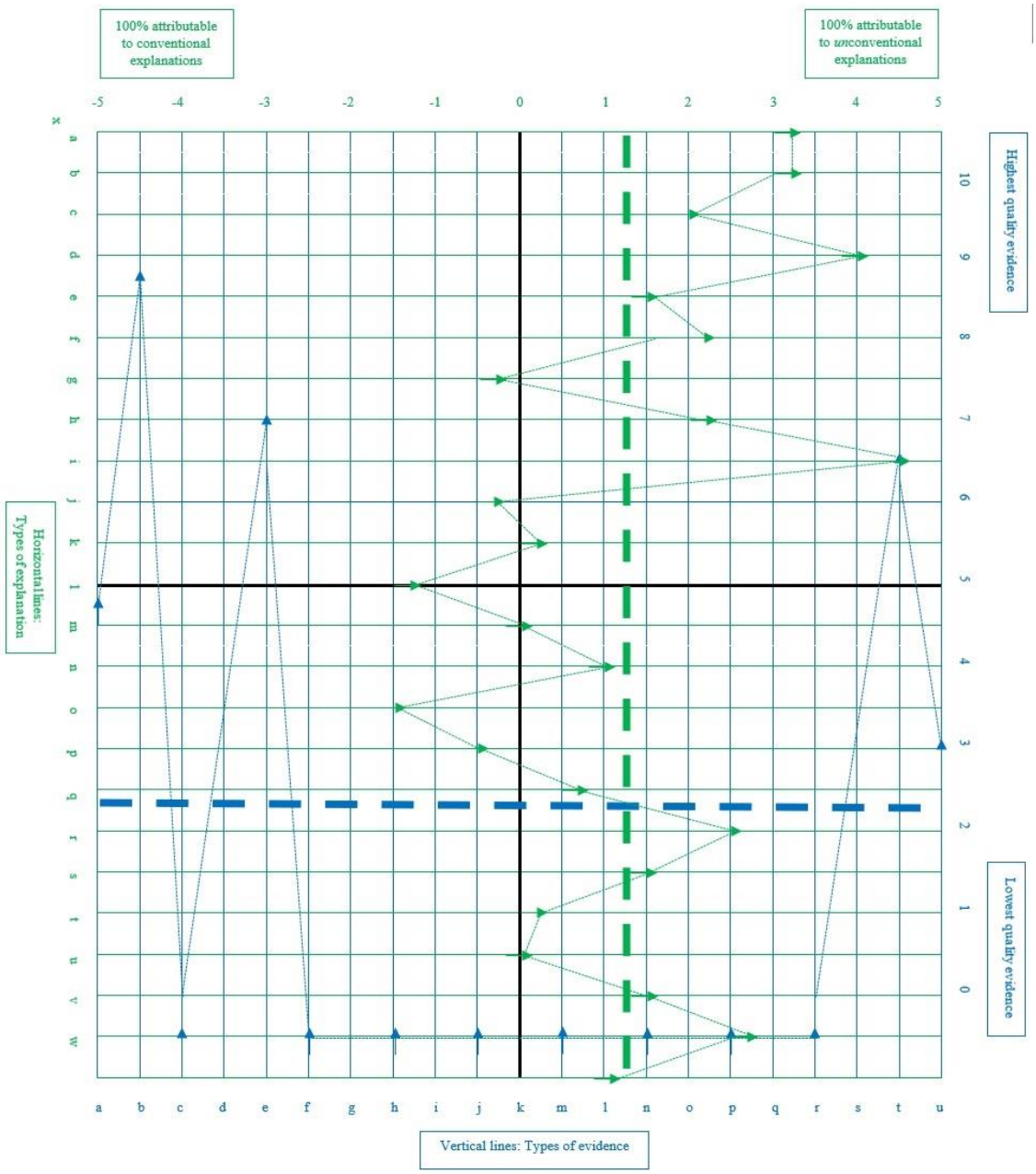
In December 1953, renowned aerospace engineer Clarence "Kelly" Johnson, Johnson's wife, and five Lockheed Aircraft Corporation pilots and engineers reported an anomalous sighting off the coast of southern California. (Note: this region has since become well-known for UAP activity, and there have even been allegations of a UAP underwater "base" there, as for example mentioned by Retired Admiral Tim Gallaudet (2023) – former head of the National Oceanic and Atmospheric Administration – who said, "You know, there's talk about, is there a UAP base in Southern California? It sounds wild, but after this week's hearing [i.e., the Congressional UAP hearing on July 26, 2023], maybe not.") The details of Johnson's sighting are contained in a collection of first-person reports, "Sighting of a flying saucer by certain Lockheed Aircraft Corporation personnel on 16 December 1953," of which an unredacted, highest-quality copy is contained in the Ben Rich collection of the Huntington Library (Lockheed Aircraft Corporation, 1954). A report of the incident is also available in the National Archives Catalog (1953), involving the US Air Force Project Blue Book case files, and on The UFO Database (no date). The incident is particularly noteworthy given the expertise of the observers. At the time, Johnson, who would later lead the design of the famed U-2 and SR-71 reconnaissance aircraft, was Lockheed Aircraft Corporation's chief engineer. Johnson, along with the five Lockheed pilots and engineers involved in the sighting, can be considered among the most knowledgeable and credible observers of aerial phenomena of their era. As noted in their accounts, Johnson and his colleagues had significant experience observing aircraft of all types. Just as importantly, the December 1953 observations occurred from two distinct vantage points. Johnson and his wife were at their ranch in Agoura, California, while the five Lockheed pilots and engineers were airborne in a WV-2 test aircraft over the Pacific Ocean in the vicinity of Long Beach, California.

In Johnson's own report (p.4 of the manuscript), the incident began at approximately 5:00 pm on Wednesday, December 16, when he observed what he "first thought was a black

cloud" in the sky. However, he "wondered why this one object was so dark, considering that the sun was behind it. I immediately thought that some aircraft had made an intense smoke trail, so I studied the object closely. It was apparent, after my first few seconds of consideration, that the outline of the object did not change… When it did not move or disintegrate," Johnson asked his wife to bring them their "eight-power binoculars, so I would not have to take my eyes off the object, which by now I had recognized as a so-called 'saucer'. As soon as I was given the glasses, I ran outside and started to focus the glasses on the object, which was now moving fast on a heading between 240° and 260° [west-southwest]… I gathered its speed was very high, because of the rate of fore-shortening of its major axis. The object, even in the glasses, appeared black and distinct, but I could make out no detail, as I was looking toward the setting sun, which was, of course, below the horizon at the time." The remaining key points about the case are summarized in Appendix 2, which gives a fuller account compiled from sources cited above. Having reflected on this case, the authors sought to apply the matrix to it. To begin, we each individually answered the various questions with respect to this case, giving our scores for all the vertical and horizontal dimensions. These individual scores are featured in appendices 3-6 (i.e., one appendix per author), together with some personal commentary and reflection on why we had selected various scores. We then created an overall shared matrix by aggregating our respective scores (i.e., by calculating the mean) for each vertical and horizontal dimension. The result is the matrix shown in Figure 3 below. In interpreting this Figure, here are the key points to bear in mind:

- The vertical dimension, which pertains to the nature of the evidence, is coloured blue
    - The blue vertical columns, labelled a - u along the bottom, refer to the types of evidence
    - The blue numbers, from 1-10 on the right-hand side, denote the quality of the evidence
    - The blue vertical arrows signify the quality of any given type of evidence
    - The thick, dotted blue horizontal line reflects the mean score of all the blue arrows
    - In this case, the blue line denotes a mean score of 2.3, meaning that overall the evidence is judged as being relatively low quality
- The horizontal dimension, which pertains to the types of explanation, is coloured green

    - The green horizontal lines, labelled a - x along the left-hand side, refer to the types of explanations.
    - The green numbers, from -5 - +5 along the top, reflect the plausibility of the various explanations in accounting for the data, where -5 means the data are 100% attributable to conventional explanation, and +5 means they 100% require unconventional explanations.
    - The green horizontal arrows signify the extent to which the different explanations are rated as plausible; in the case of conventional explanations, the more plausible it is, the closer the arrow is to -5, while the less plausible it is, the closer the arrow is to +5; conversely, in the case of unconventional explanations, the more plausible it is, the closer the arrow is to +5, while the less plausible it is, the closer the arrow is to -5.
    - In this case, the green line denotes a mean score of 1.2, meaning that overall the case is leaning toward suggesting an unconventional explanation.
- The intersection of the thick, dotted lines indicates where the case overall is situated on the matrix
    - In this case, the intersection falls within the lower right quadrant, which was described above as “low quality evidence not in conformity with current understanding.”



‘

Figure 3. Applying the UAP-AM to the case study

Conclusion

This paper has presented the first iteration of a matrix designed for appraising UAP cases, featuring two main dimensions. First, a vertical dimension of evidence (i.e., the quality of the data pertaining to a case), plotted on a scale from 0 (i.e., no evidence at all) to 10 (i.e., highest quality evidence). Second, a horizontal dimension of understanding (i.e., the extent to which the data is plausibly accounted for by various theories and explanations), plotted from -5 (i.e., the case is 100% attributable to conventional explanations) to +5 (the case 100% demands an unconventional explanation). Moreover, both dimensions feature numerous sub-dimensions (which is what makes the framework a *matrix*), allowing more nuanced and fine-grained

assessments to be made. Thus, we can firstly assess different potential types of evidence on the 0-10 scale, including eye-witness testimony, photographic data, radar data, and so on. Then, in terms of understanding, we can consider the plausibility of various explanations, including both conventional types (where the greater their plausibility, the nearer their score is to -5) and unconventional types (where the greater their plausibility, the nearer their score is to +5). Thus, the matrix allows for considerable nuance and granularity in assessing a given case. However, it also permits a broad-brush appraisal, whereby the various vertical dimensions can be aggregated into a single score of overall quality, and similarly the various horizontal dimensions aggregated into a single overall appraisal of whether the case is more likely attributable to conventional or unconventional explanations). The intersection of these two overall scores then serves to situate the case into one of four overarching quadrants: (a) high quality evidence in conformity with current understanding (upper left); (b) low quality evidence in conformity with current understanding (lower left); (c) high quality evidence not in conformity with current understanding (upper right); and (d) low quality evidence not in conformity with current understanding (lower right).

In addition to introducing the matrix, we piloted it here by applying it to a particular UAP case study, namely, a sighting in December 1953 by renowned aerospace engineer Clarence "Kelly" Johnson, together with his wife, and five colleagues, off the coast of southern California. In terms of our overall appraisal, this received a quality score of 2.3 (meaning it is of relatively low quality), and a score pertaining to explanations of 1.2 (meaning, on balance, it does not conform with current understanding, and may require extraordinary explanations). We must emphasise though that this received a score of low quality because it relies on eye-witness accounts, and lacks other kinds of data. Although the eye-witness accounts are essentially of the highest quality (indeed, it is hard to conceive of better witnesses), the case received scores of 0 for the other types of evidence, which dragged the mean down considerably. Speaking reflectively and honestly, we are unsure if this "skewing" represents a limitation/flaw with the matrix, or conversely, whether it is actually an appropriate result. On one hand, it seems unfortunate that a case that involves the highest quality eye-witness testimony should be dragged down simply because other types of data are missing. On the other hand, it may well be fair for cases relying on eye-witness testimony alone to be judged relatively poorly, leaving room for cases with multiple lines of evidence to receive far higher scores. Yet this kind of conundrum is precisely why we wish to offer this matrix as a first "iteration," with it then being subject to revision in light of feedback to the paper. This is not a finalized version, but simply our best first iteration, which we hope

people interested in the UAP topic will engage with and try out, and which we shall then refine in response to comments and suggestions.

We therefore hope the matrix is useful and generative in the way we intended, and that it will provide a foundation for more nuanced and considered appraisals of UAP going forward. In that regard, our intention would be for people interested in this topic to use the matrix for any UAP case that seems interesting and merits due consideration, such as for example the recent drone sightings on the US East Coast (and elsewhere). Eventually we could envisage the matrix being hosted on an interactive website that allows people to engage with it in a convenient way, both individually and collectively (e.g., allowing people to aggregate and combine scores). In the meantime, we would invite people just to explore it manually in a "paper and pen" type way. As people start to use the matrix, it will be valuable to see what kinds of practical considerations arise. For example, as we mentioned above, people will differ in their ability to appraise the evidence, meaning that not all attempts at the matrix will be equal: some people's takes will more accurate than others, so should be assigned greater weight in some way. For example, people could be assigned (either by themselves or by others) to categories based on their level of expertise and knowledge in relation to the various topics covered by the matrix (i.e., UAP events and history, appraisal of evidence, scientific theory, etc.), such as "beginner," "intermediate" and "advanced." In calculating a composite score, an intermediate rater's score could have twice the weight of a beginner, and an advanced rater could have triple. Say for a given parameter the beginner gave it a score of 3, the intermediate 5, and the advanced rater 7, rather than simply compute a basic mean – i.e., $(3 + 5 + 7) \div 3 = 5$ – the calculation would be weighted accordingly, i.e., $([3 \times 1] + [5 \times 2] + [7 \times 3]) \div 6 = 5.7$. One could of course make the gradings and weightings even more fine-grained, such as assigning expertise on a score of 1-10. Just as we encourage people to experiment with the matrix itself, we would also encourage testing out different weighting systems. Through such refinement, we will iterate towards the best possible version of this matrix and its application, and hopefully it can thereby offer a good tool for helping the community better understanding this vitally important topic.

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Appendices

Appendix 1: Reflections on assessing evidence

In order to establish some baselines and guidelines for assessing evidence relating to UAP, in this section we draw on and learn from frameworks and standards that some other disciplines have already developed to assess the value of pieces of evidence in their particular domain. While much has been written elsewhere about the epistemology and philosophy of evidence, the focus here will be on scientific and legal evidence in particular. In that respect, in general, evidence is considered most valuable if it is verifiable, reproducible, and objectively analyzed to contribute meaningfully to the hypothesis testing process.

In the study of UAP events, evidence most likely comprises data collected through systematic observation or experimentation; this could encompass a range of data types, including but not limited to, visual recordings, radar tracks, and eyewitness reports. The value of this evidence could be determined by its empirical quality and the rigour of the methods used in its collection and interpretation. The burden of proof in scientific inquiry requires researchers to present evidence that is not only quantitatively sufficient but also qualitatively robust. This involves a critical assessment of the source, methodology, and context of the data. Evidence that conforms to established scientific principles and is supported by high-

quality data is more likely to be integrated into the existing body of knowledge. In contrast, evidence that challenges these principles demands a higher threshold of scrutiny, requiring rigorous validation processes. In UAP research, where the phenomena under investigation often lack precedents, the interpretation of evidence must be approached with an emphasis on scientific objectivity and methodological precision. It is essential that any conclusions drawn are firmly supported by empirical evidence and are consistent with the principles of scientific inquiry.

In the field of medicine, the objective assessment of the quality of scientific evidence is critical in developing clinical guidelines and healthcare decision-making. New approaches to treatments of disease or the deployment of novel drugs and vaccines require rigorous scrutiny before being offered to patients. There are various frameworks and methods that are used to structure this objective assessment. The critical appraisal of scientific literature constitutes a fundamental aspect of research in the sciences, serving to systematically evaluate research studies to ascertain their validity, findings, and applicability in real world settings. This process of evaluation encompasses an analysis of several components of a research study, including its design, methodological approach, data gathering techniques, statistical analyses, and the pertinence and applicability of its conclusions. The primary aim of critical appraisal is to distinguish research of high quality, which provides reliable and relevant outcomes, from studies that are methodologically flawed or of limited contextual relevance. This discernment is crucial as it prevents the implementation of interventions based on evidence that is either ineffective or potentially deleterious, thereby safeguarding against the reliance on poor-quality evidence. In the domain of critical appraisal, the Joanna Briggs Institute (JBI) (2017) checklists and the Critical Appraisal Skills Programme (CASP) (2023) checklists are recognized for their efficacy. These offer a comprehensive range of appraisal instruments tailored to diverse research methodologies, encompassing randomized controlled trials, qualitative research, systematic reviews, and economic evaluations, among others. These tools facilitate a systematic inquiry into potential biases and the methodological integrity and credibility of studies. They present a structured series of inquiries that enable a thorough examination of the crucial facets of a study, focusing on its validity, outcomes, and relevance within a specific contextual framework. The utilization of these instruments enables healthcare professionals and researchers to effectively evaluate the quality of evidence, thereby informing judicious decision-making in their respective fields.

While JBI and CASP focus on the initial evaluation of individual studies, assessing methodological quality, validity, and applicability, there are other frameworks that are

employed at a subsequent stage of analysis, primarily for synthesizing and grading the overall quality of evidence across multiple studies and for developing guidelines. One common approach to developing and assessing clinical guidelines is the Grading of Recommendations Assessment, Development and Evaluation (GRADE) framework (Guyatt et al, 2008), which aims to create a structured, transparent, and informative system for rating the quality of evidence. Developed by an international group of scientists and clinicians, the GRADE methodology assesses evidence quality across studies in terms of risk of bias, inconsistency, indirectness, imprecision, and publication bias. It classifies evidence as high, moderate, low, or very low quality. High-quality evidence indicates strong confidence that the true effect lies close to the estimate, whereas very low-quality evidence signifies that the true effect is likely substantially different. GRADE also helps in formulating recommendations by considering the balance between benefits and harms, patients' values and preferences, and resource use. It is widely adopted by healthcare organizations and guideline developers worldwide, enhancing the clarity and comparability of healthcare recommendations. Several prominent organizations use the GRADE framework to develop their guidelines, reflecting its widespread acceptance and utility in the healthcare and research fields, including the World Health Organization (2014), the Cochrane Collaboration (Higgins et al, 2021), the American College of Physicians (Qaseem et al, 2010), the UK National Institute for Health and Care Excellence (2012), the Scottish Intercollegiate Guidelines Network (2019) and the Canadian Task Force on Preventive Health Care (2022).

In other fields, such as law and policing, frameworks have emerged that allow investigators to assess information for its value without seeking to define absolutely its 'truth.' In law, for example, expert witnesses are set apart from regular witnesses in that they are invited to provide an opinion in court based on their expertise and the scientific techniques they employ. Case law has established standards that must be met in order to consider an expert witness's testimony admissible in court; this is known as "The Daubert Standard" and applies in US Federal Courts, but also is applied in other jurisdictions internationally. Established by the Supreme Court in 1993 in the case of *Daubert v. Merrell Dow Pharmaceuticals, Inc.*, this standard requires that the judge act as a gatekeeper to determine whether an expert's testimony is based on scientifically valid reasoning and methodology. Under this standard, several factors are considered: whether the theory or technique can be and has been tested, whether it has been subjected to peer review and publication, its known or potential error rate, the existence and maintenance of standards controlling its operation, and whether it has attracted widespread acceptance within a relevant

scientific community. The Daubert Standard differs from the previously used "Frye Standard" (*Frye v. United States*), which focused primarily on whether the expert's methods were generally accepted in the scientific community, offering a more flexible and rigorous approach to assessing scientific testimony in court.

It should be noted that these standards used in law to ascribe credibility to an expert witness are of limited utility when considering other types of technical evidence. Typically, each item of evidence must be tested in court to establish its veracity. There is therefore no hierarchy of evidentiary quality that defines what types of evidence are more valuable than others. For example, documents are not necessarily more valuable than eyewitnesses since some documents might be falsified, and some eyewitnesses are extremely credible and corroborated. Of course, the opposite can also be true: a verified document of known provenience might be considered more credible when it contradicts a single eyewitness whom the court considers unreliable for some reason. In any case, types of evidence are not necessarily superior to others.

The International Criminal Tribunal for the Former Yugoslavia (ICTY) dealt with war crimes that took place during the conflicts in the Balkans in the 1990s; many trials at ICTY prosecuted events where witnesses, documents and other forms of evidence reflected the complexity and scale of the alleged crimes. Some Trial Chamber Judgements sought to categorise the evidence available and ascribe 'value' to it. *Prosecutor v Radovan Karadžić*, for example, described the different types of documentary evidence and held which would be admissible and why. Media statements, for example, were not admissible unless a witness was able to attest to their accuracy. Had crimes and the trial taken place 10 years later it seems likely that other types of publicly available information, such as social media posts and "open source intelligence", would also be considered in this way. Similarly, intercepted communications were only admissible if operatives or participants attested to their authenticity.

Various jurisdictions have also established factors that make eye witnesses reliable and admissible, and what can contribute to their deliberate or inadvertent misleading of the court. *Prosecutor v Kupreškić et al* overturned the trial judgement of the appellants largely on the basis of the "frailties of human perceptions" in relation to eye witness accounts and the potential for them to misidentify the accused. The appeals judgement also cites other factors that could render an eye witness less reliable, including "the passage of time between the events and the testimony of the witness, the possible influence of third persons, discrepancies, or the existence of stressful conditions at the time the events took place". In the United

Kingdom, *R. v Turnbull* held that cases involving single eyewitnesses should be withdrawn "from the jury and direct an acquittal unless there is other evidence which goes to support the correctness of the identification". *U.S. v Wade* held that 'influence of improper suggestion upon identifying witnesses probably accounts for more miscarriages of justice than any other single factor--perhaps it is responsible for more errors than all other factors combined'...And the dangers for the suspect are particularly grave when the witness' opportunity for observation was insubstantial, and thus his susceptibility to suggestion the greatest...the vagaries of eyewitness identification are well known".

Intelligence-led investigations in policing and the security services seek to overcome the challenges associated with eyewitnesses (and sources) by using frameworks to methodologically assess the value of information based on the reliability of a source and the credibility of what it is they are attesting. For example, a "good" source might be an individual who has worked with the investigator for many months and previously provided information that has eventually proven correct; credible information might be immediately corroborated by other independent sources of human, signals or open-source intelligence. The UK's College of Policing (2022) has used a numerical grading system to describe the reliability of a source: '1: Reliable,' '2: Untested,' and '3: Not Reliable.' A source is deemed 'Reliable' when they are considered competent and historically trustworthy, often including human intelligence, technical, scientific, and forensic sources, provided it passes the competence and veracity tests. 'Untested' refers to a source that hasn't been used before or whose information hasn't been substantiated, warranting caution; such sources aren't necessarily unreliable, but their information should be corroborated, as in cases where the source is unknown, like "hotline" tips or anonymous whistleblowers. Lastly, 'Not Reliable' is used when there are reasonable doubts about the source's reliability, stemming from concerns about its authenticity, trustworthiness, competence, motive, or the credibility of the technical equipment used; in such cases, information should only be acted upon after seeking corroboration. This reliability score is complemented by a second assessment of information credibility: Grade 'A' is for information known directly to the source, typically obtained first-hand, such as through personal observation. It's important to distinguish between direct observations and second-hand information. Grade 'B' is for information known indirectly but corroborated; it's not witnessed by the source but verified through independent and separate information or intelligence graded as 'A', possibly from technical sources, other intelligence, investigations, or enquiries. Grade 'C' applies to second-hand information told to the source by others, without the source having witnessed it. Grade 'D' is used when the information's

assessment is not possible, which can include anonymous tips. Lastly, Grade 'E' is for information suspected to be false, regardless of its source.

In summary, high-quality evidence, as outlined in this section, is characterized by its adherence to rigorous scientific and methodological standards across various disciplines. In the realm of scientific inquiry it demands empirical robustness, encompassing verifiable and reproducible data collected through observation or experimentation. In considering UAP reports in the scientific literature or elsewhere, high-quality evidence could be defined by its critical appraisal through established frameworks, ensuring methodological integrity and validity. In legal contexts, high-quality evidence relies on the credibility and scientific basis of expert testimony, guided by standards such as the Daubert Standard. Similarly, in intelligence and policing, it is gauged through frameworks evaluating the reliability of sources and the credibility of information. Collectively, high-quality evidence is distinguished by its empirical quality, methodological rigour, and the structured assessment of its validity, relevance, and reliability in contributing to a sound and objective understanding of the phenomena or issues under investigation. To that end, we have constructed a UAP assessment matrix that might ideally allow observers to systematically consider and appraise the quality of evidence relating to UAP, as we introduce next.

Appendix 1 References

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Appendix 2: Lockheed Case Study

As noted in the main text, in December 1953, renowned aerospace engineer Clarence "Kelly" Johnson, Johnson's wife, and five Lockheed Aircraft Corporation pilots and engineers reported an anomalous sighting off the coast of southern California. The details of this are contained in a report, "Sighting of a flying saucer by certain Lockheed Aircraft Corporation personnel on 16 December 1953," of which an unredacted, highest-quality copy is contained in the Ben Rich collection of the Huntington Library (Lockheed Aircraft Corporation, 1954). A report of the incident is also available in the National Archives Catalog (1953), involving the US Air Force Project Blue Book case files, and on The UFO Database (no date). The account below is compiled from these sources:

At approximately 5:00 pm on Wednesday, December 16, 1953, Johnson observed what he "first thought was a black cloud" in the sky. According to Johnson, "The sun had gone down and the whole western sky was gold and red, with several thin layers of clouds or haze at fairly high altitude." As Johnson recalls, he "wondered why this one object was so dark, considering that the sun was behind it. [He] immediately thought that some aircraft had made an intense smoke trail, so [he] studied the object closely. It was apparent, after [his] first few seconds of consideration, that the outline of the object did not change. Thinking it was a lenticular cloud, [he] continued to study it, but it did not move at all for three minutes."

"When it did not move or disintegrate," Johnson asked his wife to bring them their "eight-power binoculars, so [he] would not have to take my eyes off the object, which by now [he] had recognized as a so-called 'saucer'. As soon as [he] was given the glasses, [he] ran outside and started to focus the glasses on the object, which was now moving fast on a

heading between 240 and 260 [west-southwest]. When [he] got the glasses focused on the object, it was already moving behind the first layer of haze. [He] gathered its speed was very high, because of the rate of fore-shortening of its major axis. The object, even in the glasses, appeared black and distinct, but [he] could make out no detail, as [he] was looking toward the setting sun, which was, of course, below the horizon at the time."

According to Johnson, "In 90 seconds from the time it started to move, the object had completely disappeared, in a long shallow climb on the heading noted. The clouds were coming onshore, in a direction of travel opposite to that of the object. The time in which [Johnson's] wife and [Johnson] studied this object was between 5:00 and 5:05. The object, which had hovered stationary for at least three minutes, appeared to be very large but, not knowing its distance from me, [Johsnon] could not estimate its dimensions. At all times the object appeared as an ellipse, with a finess [sic] ratio of the larger axis to the minor one of about 7 or 10 to 1. [Johnson] estimated the position of the object to be roughly over Point Mugu, which lies on a bearing about 255˚ from [his] ranch." Johnson memorialized his account on December 18, 1953, two days after the incident.

R. L. Thoren, Lockheed chief flight test engineer, along with four other pilots and engineers, was airborne at the time of Johnson's sighting. While meeting with Johnson and several other Lockheed engineers the day after the incident, Thoren "casually mentioned (for fear of being ridiculed) that [he] had been chasing a flying saucer last night. Kelly [Johnson] snapped this up immediately, and said he knew exactly where it was and when; and, with no further adieu, he said it was at 5:05 and the object was sighted off of Point Mugu."
"This literally bowled [Thoren] over, because the location of the object that [he] sighted was off of Point Mugu. [Thoren] had estimated that it was somewhere between Point Mugu and the Santa Barbara Islands. Incidentally, at the time [Thoren] had sighted it, [the aircrew] were flying over the ocean just off of Long Beach.

Thoren continued, "Kelly [Johnson] then related that last night at about 5:05 p.m. he had seen an object in the western sky and had gotten binoculars and looked at it in detail. He described it as a wing with an aspect ratio of approximately seven. He said that it appeared stationary for several minutes, and then heading directly west it disappeared in one to two minutes, as [Thoren] recollect[s] his conversation. This story jibes exactly with what [the WV-2 aircrew] saw in flight at the same time."

Describing the WV-2 test flight, Thoren stated that on "Wednesday, December 16, 1953, [he] participated in a test flight of a Navy Super Constellation WV-2, taking off at 4:29 p.m. The flight consisted of: Roy Wimmer, pilot; [Thoren], co-pilot; Charlie Grugan, flight

engineer; and Joe Ware, flight test engineer." P. A. Colman, Lockheed's then-chief aerodynamics engineer, was also aboard the WV-2.

After running some tests, the WV-2 began a climb to 20,000 feet. According to Thoren, "Somewhere between 15,000 and 20,000 feet, Roy Wimmer said to [Thoren], 'Look out, there's a flying saucer.' [Thoren] looked out the windshield towards where Roy was pointing and saw some sort of an object at approximately the altitude that [the aircrew] were flying. [Thoren] made a slight turn heading right towards the object, expecting to overtake it so that [the crew] could look at it more closely. [Thoren] maintained this heading for roughly five minutes, looking at the object all the time."

As Thoren notes, "Wimmer, Ware and [Thoren] viewed this thing for at least five minutes, discussing what [they] thought it might be. Wimmer's first impression was that it was a small cloud. After studying it for several minutes though, [Thoren] deduced that it was not a cloud because it had too definite sharp edges and its appearance stayed constant. It looked to [Thoren] like [he] was flying right directly towards, and at about the same elevation as, a very large flying wing airplane."

According to Thoren, "Although the object appeared to be absolutely stationary, [the WV-2 aircrew] did not seem to be closing the gap between [them] and this object, even though [they] were flying at some 225 miles per hour. The object then seemed to be getting smaller, and [Thoren's] attention was diverted from it for a minute or so, but Wimmer mentioned that the object was disappearing. In probably an elapsed time of somewhere around a minute, the object had reduced in size to a mere speck, and then disappeared. Its direction was almost due west. At the same time, the sun had gone down below the horizon but the sky was red, and this object silhouetted perfectly against this red background. The atmosphere was extremely clear."

"When [Thoren] first sighted the object, [he] guessed that it was probably seven miles away. However, looking at it in retrospect, the object must have been considerably larger than [he] had estimated and, hence, the distance was probably much greater than [he] had also estimated."

According to Thoren, he was "very skeptical of flying saucer stories, and [he had] never even imagined seeing an object in the sky that [he] was not able to identify. The three of [them] who watched it from the airplane are all pilots who have been flying for many years on experimental test work, and are trained to have accurate observations. Kelly [Johnson] also has had a lot of experience in flight test work and has been flying for many years and is also a very trained observer. The fact that what he saw and what [the WV-2 aircrew] saw

appears to be identical, and the time and place identical, leads [Thoren] to believe that it was not exactly an illusion that [he] observed." Thoren memorialized his account on December 17, 1953, one day after the incident.

Other aircrew accounts are largely identical to Thoren's. As pilot Roy Wimmer recounts, "After watching it for a few minutes [the aircrew] decided that it wasn't a cloud but some kind of object. It had a definite shape which appeared to [Wimmer] like a crescent. Others on board described it as a huge flying wing. [Wimmer] could not detect any details other than the shape of it. [Wimmer] estimated the distance from [the WV-2] to be at least fifty or sixty miles and possibly much further. In the clear air like that it is very hard to judge distance. [The WV-2 crew] flew directly toward it for about five minutes and [their] relative position did not appear to change."

With Thoren flying the WV-2, Wimmer "had nothing else to do but to watch the object. After about five minutes [Wimmer] suddenly realized it was moving away from [them] heading straight west. In the space of about one minute it grew smaller and disappeared. [Wimmer] was watching it all the time so [he] was able to see it for several seconds after the rest of the crew lost sight of it. Right up until the time it disappeared it maintained its sharp outline and definite shape so [he knew] it was not a cloud that dissolved giving the appearance of moving away."

Wimmer "[has] had considerable experience, while doing radar bombing on P2V's, of estimating distance where there is very little to judge by and [he is] convinced this was a large object some distance away." Wimmer wrote his account on January 12, 1954, nearly a month after the sighting.

P.A. Colman, Lockheed Aircraft Corporation's chief aerodynamics engineer, was also aboard the WV-2. According to Colman, "While flying off the coast in the vicinity of Santa Monica, [he] saw an object apparently standing still in the air off the coast, in the vicinity of Point Mugu… The object appeared as a thin black line, giving a first reaction of a B-36 type airplane, heading straight toward [the WV-2 aircrew] and silhouetted against a bright background. The background was bright due to the fact that the sun was just setting. The object appeared not to move while [the aircrew] progressed with [their] tests. For a few moments [they] turned the airplane toward the object but did not apparently change [their] distance sufficiently to get any change of impression. [Colman] estimate[s] that the object was hovering in [their] sight for about ten minutes. Thereafter, it suddenly accelerated due west and in a time, in the order of 10 seconds, disappeared from view."

As Colman recalls, the "following day it was revealed that Mr. Clarence L. Johnson had seen the identical object while standing on the ground at his ranch. This coincidence is interesting. The difference in the positions, both horizontally and vertically between us indicate that the object had sufficient depth to eliminate the possibility that it was a cloud phenomena. The similarity of the explanations of the shape and actions of the object is remarkable. However, the blackness made it impossible to discern anything but the basic outline." Colman's account is dated January 12, 1954, roughly a month after the incident.

J. F. Ware, Lockheed flight test supervisor aboard the WV-2, recalls that the phenomenon "seemed to be stationary, although [the aircrew] did not appear to overtake it at all. [Ware's] first thought was that it was a large airplane, possibly a C-124, but after looking more closely, it seemed to look more like a large object without wings with a maximum thickness in the middle tapering toward either side. [Ware] could not distinguish front or rear on the object. It seemed to be somewhat above [the WV-2] and to the West, over the water, possibly in the vicinity of Santa Barbara Islands."

Ware continued, "After looking at the object off and on for about five minutes, it became apparent that it was moving away from [the WV-2 aircrew] and in just a minute or two it completely disappeared. As it was disappearing, [Ware] looked at it off and on and gradually [he] could not see it at all. Roy [Wimmer] watched it continuously and could see it after [Ware] had lost sight of it – [Wimmer] actually observed it continuously… It disappeared in a generally westward direction (toward the setting sun)." Ware wrote his report on January 11, 1954, nearly a month after the incident.

According to Lockheed chief engineer Johnson, two years before the December 1953 sighting, he and his wife "saw an object which [he] believed at the time, and still do, to be a saucer." Johnson closed his report of the December 1953 incident by stating that "for at least five years [he has] definitely believed in the possibility that flying saucers exist – this in spite of a good deal of kidding from [Johnson's] technical associates. Having seen this particular object on December 16th, [Johnson is] now more firmly convinced than ever that such devices exist, and [he has] some highly technical converts in this belief as of that date."

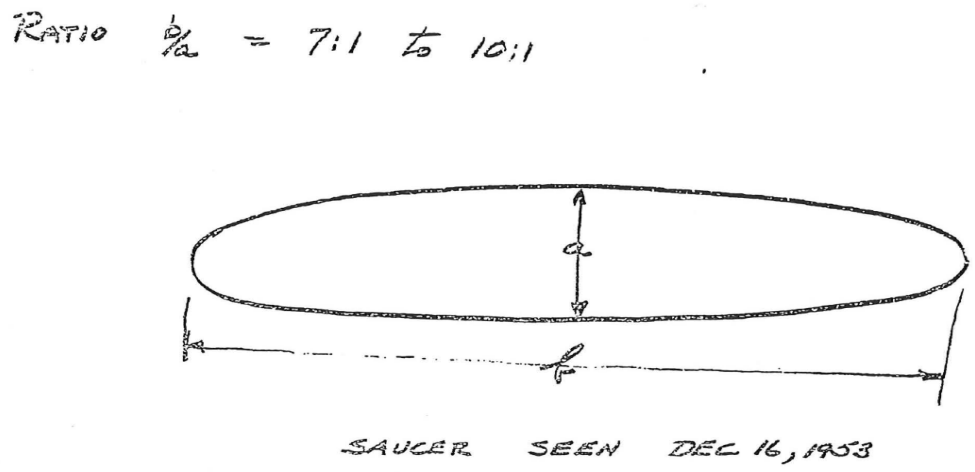


Figure 4: Lockheed chief engineer Clarence “Kelly” Johnson’s sketch of the UAP

Appendix 3: M.M. annotated responses to questionnaire

Section 1 (quality of evidence)

- Va. Amount of direct witnesses - 3
- Vb. Quality of direct witnesses - 9
    - Rating of 9 given rather than 10 because some of the seven direct witnesses included Kelly Johnson’s wife who isn’t highly trained and as experienced as

the other 6, and because direct witness J.F. Ware only looked at the object 'off and on' and he lost sight of it before the others.

- Vc. Number of secondary witnesses (e.g., saw radar footage or other data) – 0
- Vd. Quality of secondary witnesses – N.A.
- Ve. Triangulation (i.e., "agreement") among witnesses - 8
- Vf. Existence of photographic data – 0
- Vg. Quality of photographic data – N/A
- Vh. Existence of video data - 0Vi. Quality of video data – N/A
- Vj. Existence of radar data – 0
- Vk. Quality of radar data – N/A
- Vl. Existence of infrared data (e.g., FLIR) – 0
- Vm. Quality of infrared data – N/A
- Vn. Existence of other electromagnetic signals (i.e., other than infrared and visible light) – 0
- Vo. Quality of other electromagnetic signals – N/A
- Vp. Existence of physical/material evidence (i.e., recovered material from UAP) – 0
- Vq. Quality of physical/material evidence – N/A
- Vr. Number of other data sources (i.e., other than those listed above) – 0
- Vs. Quality of other data sources – N/A
- Vt. Triangulation (i.e., "agreement") among data sources – 7
- Vu. Public availability of data - 4

Section 2a (conventional theories)

- Ha. A physical psy-op (i.e., an incident genuinely occurred, but was staged by humans, intended to mislead observers into believing it is a 'real', i.e., extraordinary, UAP
    - 8 – Considering the time period and knowledge of prosaic technologies available to the US military at that time it is highly unlikely but not impossible this was staged by the engineers' contemporaries.
- Hb. A hoax or lie (i.e., reports and evidence of the event are fabricated
    - 9 – Since multiple different highly skilled engineers observed the object it is unlikely this was a coordinated lie amongst them, especially since they were subjected to ridicule by their peers so it doesn't make sense why they would subject themselves to that.

- Hc. Misperception (i.e., genuine physical stimuli were observed, but were misinterpreted)
    - 7 – The engineers and pilots who observed the craft were all highly trained and attempted to find a more prosaic explanation for what they were observing, but ultimately concluded that it was a disc-shaped UFO base on the fact that it was solid (not a cloud) and capable of hovering and moving at a high rate of speed.
- Hd. Hallucination (i.e., genuine physical stimuli were *not* observed, but were merely imagined)
    - 9 – It is highly unlikely that each of the 7 people who witnessed the object, both from the ground and at altitude were hallucinating.
- He. Conventional airborne objects (e.g., a balloon)
    - 8 – The weather balloon explanation was a very common one at the time this object was reported but given the size of the craft and its ability to hover in place and move rapidly, in addition to the fact that it was a solid shape and didn't change in association with the wind indicates it was not a conventional airborne object.
- Hf. Conventional human aviation technology (e.g., a plane)
    - 8 – Again, considering the knowledge and experience of the witnesses involved, it is unlikely they would mistake such a machine for a plane or some other conventional aviation tech, especially because Johnson was able to get a better look at it through binoculars and the other men observing from the sky also got a good and prolonged look at the object.
- Hg. Secret human aviation technology (e.g., a "black" military special access project)
    - 5 – Because the witnesses are tasked with testing the most cutting-edge aircraft available to the US military it is hard to imagine they wouldn't know about a high-tech antigravity craft like this. However, it is possible the US military could have reverse engineered a disc like that which crashed near Roswell, NM in 1947 by the time these individuals witnessed the object, and if they were outside the special access project, they wouldn't have knowledge of it.
- Hh. Explainable meteorological phenomena (i.e., a weather process)

    - 9 – Again, considering the knowledge and experience of the witnesses involved, it is unlikely they would mistake such a machine for a known meteorological phenomenon.
- Hi. Explainable biological phenomena (i.e., an airborne organism such as a bird)
    - 9 – I am unaware of any airborne organism that would fit the description provided by the witnesses, especially considering it hovered in place for some time, and because Kelly Johnson was able to observe it through binoculars.

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Section 2b (unconventional theories)

- Hj. Extraterrestrial intelligences (i.e., entities from elsewhere in the cosmos) + 2
    - The object had an unconventional solid form, hovered stationary for at least three minutes, then moved at high speed, as observed by Johnson and his wife while looking through his binoculars on the ground, and concomitantly, by R. L. Thoren and his airborne crew of four pilots and engineers while flying at near the same altitude. Such an object, and its ability to hover and move faster than typical known modern aircraft, could potentially have been made by an advanced extraterrestrial civilization.
- Hk. Extratempestrial intelligences (i.e., future human time travellers) + 4
    - The object had an unconventional solid form, hovered stationary for at least three minutes, then moved at high speed, as observed by Johnson and his wife while looking through his binoculars on the ground, and concomitantly, by R. L. Thoren and his airborne crew of four pilots and engineers while flying at near the same altitude. Such an object, and its ability to hover and move faster than typical known modern aircraft, is arguably more likely to have been made by an advanced future human population. Comparatively, this may exist as an Occam's razor interpretation since it contains fewer assumptions, especially since we know humans are here, we've had a long evolutionary history on this planet, and when considering a 3-million-year trend toward accelerating the rate of our cultural and technological evolution, we would expect future human groups to possess more advanced technology that may be similar in form and functionality to that observed by Johnson, Thoren, and the others involved in this case. In fairness, and despite an awareness of potential biases, it should be acknowledged that this rating was made by an individual who has

researched and written about this time-traveling-humans potentiality for over a decade, which may have contributed to a higher rating for this explanatory model over certain others.

- Hl. Cryptoterrestrial intelligences (i.e., entities concealed within or near Earth) – 1
    - It is unlikely that an advanced intelligence living underground would be capable of finding the requisite materials to construct, or that they would be able to test and fly a large and highly advanced craft such as that seen by the observers in this case. However, a higher rating than -5 was given because it is possible that an advanced extraterrestrial or extratempestrial cryptoterrestrial civilization could have, or will at some point in the future, develop such technology elsewhere and/or elsewhen, and conceal it in our times, either below the Earth's surface, under our vast oceans, or perhaps even on the far side of the moon.
- Hm. Interdimensional intelligences (i.e., inhabiting other dimensions beyond the four of spacetime) + 2
    - This rater considers the interdimensional theory to be a broader category of the extratempestrial model. However, it is rated slightly lower than the above because it doesn't pinpoint the actual origins of those piloting the observed craft, but rather, exists as something of an ambiguous catchall buzzword that lacks specificity. Additionally, as it has yet to be demonstrated that other dimensions exist beyond our four perceived dimensions, it is currently unknown how interdimensional travel might take place. Additionally, it is hard to fathom how something constructed in a universe with additional dimensions might be able to properly function in one with fewer dimensions.
- Hn. Angelic intelligences (i.e., spiritual entities, as described in religious teachings) + 1
    - It is possible that the observed object could have been classified as an angelic intelligence depending on the specific era when it was observed and the belief system of the observers. It is likely that this idea of angelic intelligences, and arguably most global religions, grew out of observations like this. However, the concept of an angelic intelligence is seen here as more of an interpretation of observations of physical craft and their pilots, rather than as an explanation for their origins.

- Ho. Unexplained meteorological phenomena (i.e., a weather process humans are unaware of) – 3
    - Considering the high level of experience, expertise, and the credentials of the witnesses in this case, both on the ground while looking through binoculars and while observing the object in the sky at the same relative altitude, it is highly unlikely that it represented some form of prosaic meteorological phenomena. However, because each witness's first thought was that it may be a cloud, this potential explanation should be acknowledged.

Section 2c (current scientific knowledge)

- Hp. Physics – 7
- Hq. Material science and technology – 6
- Hr. Aviation science and technology – 6
- Hs. Biology - 9
- Ht. Chemistry - 9
- Hu. Psychology - 8
- Hv. Anthropology - 7
- Hw. Economics - 7
- Hx. Politics - 6

Appendix 4: T.L. annotated responses to questionnaire

Section 1

- Va. Amount of direct witnesses: 5
    - There were 7 witnesses, which on a 1-10 scale I would rate as about a 5 (i.e., 7 is a decent amount, but we need to leave scope for much higher scores for events that might have dozens and even hundreds)
- Vb. Quality of direct witnesses: 8
    - Some of the witnesses are about as high a quality as one can get (e.g., I would give Kelly Johnson himself a score of 10), but I'm not confident in the nature/skills of all the other witnesses (though presumably they would generally score pretty highly), hence giving an 8 overall.
- Vc. Number of secondary witnesses (e.g., saw radar footage or other data). 0
    - As far as I can tell, there are no secondary witnesses, which I think should make this an automatic 0
- Vd. Quality of secondary witnesses. N/A
    - Given the lack of secondary witnesses, I suppose this answer should also technically be 0. However, that feels like a kind of "double jeopardy"! Maybe we should introduce a stipulation that if a case receives a 0 in relation to not having a type of data (and this applies to the forms listed below), then we shouldn't also give a 0 in relation to quality, but simply instruct people to answer N/A to the quality?
- Ve. Triangulation (i.e., "agreement") among witnesses: 7
    - The accounts of the various people involved did seem to match up quite well and to some extent corroborate each other.
- Vf. Availability of photographic data: 0
    - For the reason stated in relation to the number of other data sources.
- Vg. Quality of photographic data: N/A
    - For the reason stated in relation to the quality of secondary witnesses.
- Vh. Availability of video data: 0
    - For the reason stated in relation to the number of other data sources.
- Vi. Quality of video data: N/A
    - For the reason stated in relation to the quality of secondary witnesses.
- Vj. Availability of radar data: 0
    - For the reason stated in relation to the number of other data sources.
- Vk. Quality of radar data: N/A
    - For the reason stated in relation to the quality of secondary witnesses.

- Vl. Availability of infrared data (e.g., FLIR): 0
    - For the reason stated in relation to the number of other data sources.
- Vm. Quality of infrared data: N/A
    - For the reason stated in relation to the quality of secondary witnesses.
- Vn. Availability of material/biological data (i.e., recovered material from UAP): 0
    - For the reason stated in relation to the number of other data sources.
- Vo. Quality of other electromagnetic signals – N/A
- Vp. Existence of physical/material evidence (i.e., recovered material from UAP) – 0
- Vq. Quality of physical/material evidence – N/A
- Vr. Number of other data sources (i.e., other than those listed above) – 0
- Vs. Quality of other data sources – N/A
- Vt. Triangulation (i.e., "agreement") among data sources – 7
- Vu. Public availability of data - 3

Section 2a

- Ha. A physical psy-op (i.e., an incident genuinely occurred, but was staged by humans, intended to mislead observers into believing it is a 'real', i.e., extraordinary, UAP)
    - 9. I'm fairly skeptical about psy-op theories anyway, and in this case think it nearly impossible, based on the quality of the witnesses (i.e., I don't see how Johnson would fall for it).
- Hb. A hoax or lie (i.e., reports and evidence of the event are fabricated)
    - 9. Again, I just don't see any reason why Johnson and colleagues would lie about this, especially given the ridicule they faced from peers.
- Hc. Misperception (i.e., genuine physical stimuli were observed, but were misinterpreted).
    - 8. This is always a fairly high probability, but the quality of the witnesses makes it very unlikely in this particular case.
- Hd. Hallucination (i.e., genuine physical stimuli were *not* observed, but were merely imagined).
    - 9. Hallucinations are rather rare, and shared hallucinations (i.e., multiple people) almost never occur.
- He. Conventional airborne objects (e.g., a balloon).

    - 8. For the same reason as misperception.
- Hf. Conventional human aviation technology (e.g., a plane).
    - 8. For the same reason as misperception.
- Hg. Secret human aviation technology (e.g., a "black" military special access project).
    - 6. I suppose that, even for a person of Johnson's standing, there could be aviation projects he was unaware of, whether by the US or other nations.
- Hh. Explainable meteorological phenomena (i.e., a weather process).
    - 8. For the same reason as misperception.
- Hi. Explainable biological phenomena (i.e., an airborne organism such as a bird).
    - 9. The observations do not seem to match, in any way, biological phenomena that we are aware of.

Section 2b

- Hj. Extraterrestrial NHIs (i.e., entities from elsewhere in the cosmos)
    - 7. There doesn't seem to be many details in the present case to differentiate between different extraordinary hypotheses (other than perhaps downgrading the likelihood of the angelic one, given the craft-like nature), so my ratings here are more based on my opinions on the viability of these various hypotheses in general. In that respect, this for me is still the most likely one of all of these.
- Hk. Extratempestrial NHIs (i.e., future human time travellers)
    - 5. Am in principle very open to this hypothesis, but rate it slightly lower (in general) than the first one. Plus, I associate this hypothesis more with specific types of UAP events that seem to relate to what future humans might be concerned with (e.g., nuclear activity), and I'm not sure such considerations apply to this instance.
- Hl. Cryptoterrestrial NHIs (i.e., entities concealed within or near Earth)
    - 5. Again, also in principle open to this, but rate it roughly equivalent to the one above.
- Hm. Interdimensional NHIs (i.e., inhabiting other dimensions beyond the four of spacetime).
    - 6. For me, this kind of sits between the first two hypotheses in terms of likelihood, though to be honest, as Mike has argued, they may well all overlap

anyway (i.e., extraterrestrial travel may well involve interacting with or harnessing other dimensions in some way).

- Hn. Angelic NHIs (i.e., spiritual entities, as described in religious teachings)
    - 3. I'm open to this hypothesis in principle, and believe it may well explain some UAP (especially ones involving orbs of light), but not this particular one, given it's craft-like nature
- Ho. 'Unexplained' meteorological phenomena (i.e., a weather process humans are unaware of)
    - 6. I do think that, even now, there are lots of strange meteorological processes that we don't understand very well, and which even someone like Johnson could be surprised or misled by.

Section 2c

- Hp. Physics
    - 2. To me, what is significant about this case is that the craft does not seem to align with aviation technology (at that time) - which Johnson would be eminently placed to assess - which is what makes this an extraordinary UAP. However, apart from that third category (aviation science and technology), nothing in the available reporting seems to suggest the behaviour of the craft was contravening current scientific knowledge (unlike, for instance, the Nimitz tic-tac). That doesn't mean the case isn't extraordinary, but just that its extraordinary nature is not derived from the considerations in this section.
- Hq. Material science and technology
    - 2. For the same answer as for physics.
- Hr. Aviation science and technology
    - 9. This is the exception in this category. Clearly, Johnson thought the UAP did not conform to current aviation science and technology, hence him assessing it as a genuine UAP.
- Hs. Biology
    - 2. For the same answer as for physics.
- Ht. Chemistry
    - 2. For the same answer as for physics.
- Hu. Psychology

    - 2. For the same answer as for physics.
- Hv. Anthropology
    - 2. For the same answer as for physics.
- Hw. Economics
    - 6. These last two considerations are a little different from the rest. In theory the development of the kind of craft in the report is consistent with economic science (i.e., it would be possible to finance such a vehicle). However, Johnson's supposition is that such a craft hadn't been financed, which takes this answer into the "extraordinary" territory.
- Hx. Politics
    - 6. For the same answer as for economics.

Appendix 5: A.O. annotated responses to questionnaire

- Va. Amount of direct witnesses:
    - 5/10 - a small group
- Vb. Quality of direct witnesses
    - 10/10 - I can't think of higher quality witnesses
- Vc. Number of secondary witnesses (e.g., saw radar footage or other data).
    - 0/10 - none that we know of
- Vd. Quality of secondary witnesses. n/a
- Ve. Triangulation (i.e., "agreement") among witnesses:
    - 7/10 - some minor disagreement RE timelines, position
- Vf. Availability of photographic data: 0/10
- Vg. Quality of photographic data: n/a
- Vh. Availability of video data: 0/10
- Vi. Quality of video data: n/a
- Vj. Availability of radar data: 0/10
- Vk. Quality of radar data: n/a
- Vl. Availability of infrared data (e.g., FLIR): 0/10
- Vm. Quality of infrared data: n/a
- Vn. Availability of material/biological data (i.e., recovered material from UAP): 0/10
- Vo. Quality of material/biological data: n/a
- Vt. Triangulation (i.e., "agreement") among data source: n/a
- Vu. Public availability of data: n/a

Section 2a

- Ha. A physical psy-op (i.e., an incident genuinely occurred, but was staged by humans, intended to mislead observers into believing it is a 'real', i.e., extraordinary, UAP):
    - +3 - I think this is unlikely
- Hb. A hoax or lie (i.e., reports and evidence of the event are fabricated):
    - +1 - there is no data, only witness testimony albeit from several people in agreement with each other; this is more likely but I don't know why they would lie.

- Hc. Misperception (i.e., genuine physical stimuli were observed, but were misinterpreted):
    - +2 - several expert witnesses all reporting similar things
- Hd. Hallucination (i.e., genuine physical stimuli were *not* observed, but were merely imagined).
    - +3 - I think this is unlikely due to the number of contemporaneous witnesses in different locations
- He. Conventional airborne objects (e.g., a balloon).
    - -1 - this seems possible; the witnesses even mentioned similar aircraft that resembled whatever they saw
- Hf. Conventional human aviation technology (e.g., a plane):
    - -1 - unsure how this is different to the above
- Hg. Secret human aviation technology (e.g., a "black" military special access project):
    - -3 "flying wings" were in development around this time (YB-49 was retired but were there others)?
- Hh. Explainable meteorological phenomena (i.e., a weather process):
    - -1 it seems the observers all gave accounts whereby they suggested this might've been a cloud, but dismissed it because for various reasons.
- Hi. Explainable biological phenomena (i.e., an airborne organism such as a bird).
    - +5 too big

Section 2b

- Hj. Extraterrestrial NHIs (i.e., entities from elsewhere in the cosmos): -4
- Hk. Extratempestrial NHIs (i.e., future human time travellers) -4
- Hl. Cryptoterrestrial NHIs (i.e., entities concealed within or near Earth) -5
- Hm. Interdimensional NHIs (i.e., inhabiting other dimensions beyond the four of spacetime). -4
- Hn. Angelic NHIs (i.e., spiritual entities, as described in religious teachings) -5

Section 2c

- Hp. Physics: -3
- Hq. Material science and technology: +3

- Hr. Aviation science and technology: +3
    - because the witnesses were subject experts and couldn't explain what they'd seen
- Hs. Biology: +2
- Ht. Chemistry: -3
- Hu. Psychology: -1
- Hv. Anthropology: +4
- Hw. Economics: +4
- Hx. Politics: +4

Appendix 6: R.V. annotated responses to questionnaire

- Va. Amount of direct witnesses

    - 6 - You know for certain there are a few direct witnesses
- Vb. Quality of direct witnesses
    - 8 – military pilots and aerospace engineers
- Vc. Amount of secondary witnesses
    - N/A - We don't know whether there are secondary witnesses
- Vd. Quality of secondary witnesses
    - N/A - We don't know whether there are secondary witnesses
- Ve. Triangulation (i.e., “agreement”) among witnesses
    - 7 - The witnesses are broadly in agreement but have slight difference in a few respects
- Vf. Existence of photographic data
    - N/A - We don't know whether photographic data exists
- Vg. Quality of photographic data
    - N/A - We don't know whether photographic data exists
- Vh. Existence of video data
    - N/A - We don't know whether photographic video data exists
- Vi. Quality of video data

    - N/A - We don't know whether photographic video data exists
- Vj. Existence of radar data
    - N/A - We don't know whether photographic radar data exists
- Vk. Quality of radar data
    - N/A - We don't know whether photographic radar data exists
- Vl. Existence of infrared data (e.g., FLIR)
    - N/A - We don't know whether infrared data exists
- Vm. Quality of infrared data
    - N/A - We don't know whether infrared data exists
- Vn. Existence of other electromagnetic signals (i.e., other than infrared and visible light)
    - N/A - We don't know whether electromagnetic signal exists
- Vo. Quality of other electromagnetic signals
    - N/A - We don't know whether electromagnetic signal exists
- Vp. Existence of physical/material evidence (i.e., recovered material from UAP)
    - N/A - We don't know whether physical/material evidence exists
- Vq. Quality of physical/material evidence
    - N/A - We don't know whether physical/material evidence exists
- Vr. Existence of other data sources (i.e., other than those listed above)
    - N/A - We don't know whether other data sources exists
- Vs. Quality of other data sources
    - N/A - We don't know whether other data sources exists
- Vt. Triangulation (i.e., "agreement") among data sources
    - N/A - We don't know whether Triangulation among data sources exists

- Ha. A physical psy-op (i.e., an incident genuinely occurred, but was staged by humans, intended to mislead observers into believing it is a 'real', i.e., extraordinary, UAP)
    - +3 = 60% confident that the case is impossible to account for by the particular explanation.
- Hb. A hoax or lie (i.e., reports and evidence of the event are fabricated)
    - +4 = 80% confident that the case is impossible to account for by the particular explanation.

- Hc. Misperception (i.e., genuine physical stimuli were observed, but were misinterpreted)
    - +1 = 20% confident that the case is impossible to account for by the particular explanation.
- Hd. Hallucination (i.e., genuine physical stimuli were not observed, but were merely imagined)
    - +5 = 100% confident that the case is impossible to account for by the particular explanation.
- He. Conventional human aviation technology (e.g., a plane)
    - +2 = 40% confident that the case is impossible to account for by the particular explanation.
- Hf. Conventional airborne objects (i.e., something human-made that can be airborne but which we would not define as aviation technology, such as a "party balloon")
    - +4 = 80% confident that the case is impossible to account for by the particular explanation.
- Hg. Secret human aviation technology (e.g., a "black" military special access project)
    - +1 = 20% confident that the case is impossible to account for by the particular explanation.
- Hh. Explainable meteorological phenomena (i.e., a weather process)
    - +3 = 60% confident that the case is impossible to account for by the particular explanation.
- Hi. Explainable biological phenomena (i.e., an airborne organism such as a bird)
    - +5 = 100% confident that the case is impossible to account for by the particular explanation.

- Hj. Extraterrestrial NHIs (i.e., entities from elsewhere in the cosmos)
    - +1 = 20% confident that this hypothesis is completely untenable and invalid for the case in question
- Hk. Extratempestrial intelligences (i.e., future human time travellers)
    - +1 = 20% confident this hypothesis is unquestionably the correct one for the case in question
- Hl. Cryptoterrestrial NHIs (i.e., entities concealed within or near Earth)

    - +1 = 20% confident this hypothesis is unquestionably the correct one for the case in question
- Hm. Interdimensional NHIs (i.e., inhabiting other dimensions beyond the four of spacetime)
    - +1 = 20% confident this hypothesis is unquestionably the correct one for the case in question
- Hn. Angelic NHIs (i.e., spiritual entities, as described in religious teachings)
    - +1 = 20% confident that this hypothesis is completely untenable and invalid for the case in question
- Ho. Unexplained meteorological phenomena (i.e., a weather process humans are unaware of)
    - +2 = 40% confident that this hypothesis is completely untenable and invalid for the case in question

- Hp. Physics
    - +2 = 40% confident the case in question is impossible to explain by current scientific theory and knowledge in this particular field
- Hq. Material science and technology
    - +2 = 40% confident the case in question is impossible to explain by current scientific theory and knowledge in this particular field
- Hr. Aviation science and technology
    - +2 = 60% confident the case in question is impossible to explain by current scientific theory and knowledge in this particular field
- Hs. Biology
    - +3 = 60% confident the case in question is impossible to explain by current scientific theory and knowledge in this particular field
- Ht. Chemistry
    - +3 = 60% confident the case in question is impossible to explain by current scientific theory and knowledge in this particular field
- Hu. Psychology
    - +1 = 20% confident the case in question is impossible to explain by current scientific theory and knowledge in this particular field
- Hv. Anthropology

    - +3 = 60% confident the case in question is impossible to explain by current scientific theory and knowledge in this particular field
- Hw. Economics
    - +4 = 80% confident the case in question is impossible to explain by current scientific theory and knowledge in this particular field
- Hx. Politics
    - +1 = 20% confident the case in question is impossible to explain by current scientific theory and knowledge in this particular field

Appendix 7: Full UAP hypothesis taxonomy (Karl Nell)

Karl Nell's classificatory taxonomy has – at the highest level of abstraction – three overarching categories: (1) physical; (2) psychosocial; and (3) meta-physical. Let's take each in turn. The physical itself is differentiated into three sub-categories: (1.1) intra-dimensional; (1.2) extra-dimensional; and (1.3) inter-dimensional. These are then subdivided further. The intra-dimensional breaks down into three different types: (1.1.1) prosaic; (1.1.2) cryptoterrestrial; and (1.1.3) extra-terrestrial. Of these, prosaic has two main types of hypotheses, each with four possibilities: (1.1.1.1) natural, including (1.1.1.1.1) optical artefact / illusion, (1.1.1.1.2) celestial, (1.1.1.1.3) meteorological, and (1.1.1.1.4) geophysical; and (1.1.1.2) human, including (1.1.1.2.1) governmental, (1.1.1.2.2.) extra-governmental, (1.1.1.2.3) inter-governmental, and (1.1.1.2.4) private / commercial. Then, the cryptoterrestrial has four main hypotheses: (1.1.2.1) ancient / proto-human survival; (1.1.2.2.)

breakaway civilization; (1.1.2.3) cryptids / humanoids; and (1.1.2.4) stranded "Gods." Finally, within the intra-dimensional category, there are four extra-terrestrial hypotheses: (1.1.3.1) solar system; (1.1.3.2) inter-stellar; (1.1.3.3) inter-galactic; and (1.1.3.4) unknown region of our space-time.

The second physical sub-category is extra-dimensional, which is differentiated into two main branches: (1.2.1) 3-D entity from hyperspace, involving three hypotheses, including (1.2.1.1) "compactified" dimensions, (1.2.1.2) hyperspace "bulk," and (1.2.1.3) disconnected hyperspace; and (1.2.2) hyper-D "brane," which itself two further subdivides into two branches, the first being (1.2.2.1) projection / "shadow," which includes (1.2.2.1.1) 0-to-2 dimensional, (1.2.2.1.2.) 3-dimensional, and (1.2.2.1.3) fractal dimensional, while the second branch is (1.2.2.2) 3-D avatar, which includes (1.2.2.2.1) hologram / "telepresence," (1.2.2.2.2) automaton, and (1.2.2.2.3) artificial lifeform.

The third and final physical category, inter-dimensional, has three sub-categories: (1.3.1) multiverse, including (1.3.1.1) portal / wormhole, (1.3.1.2) colliding universe(s), (1.3.1.3) parallel universe(s), and (1.3.1.4) QFT "many worlds"; (1.3.2) time travel, which includes (1.3.2.1) humans from future, (1.3.2.2) humans from past, (1.3.2.3) humans from alternate timeline, (1.3.2.4) non-human entities, and (1.3.2.5) additional time dimensions(s); and (1.3.3) scale invariance, including (1.3.3.1) macro quantum fluctuation, (1.3.3.2) distance recursion, (1.3.3.3) pan-galactic entanglement, (1.3.3.4) unobserved "vibratory modes," and (1.3.3.5) conformal cyclic cosmology.

Turning now to the second overarching category, psychosocial, this has three sub-categories: (2.1) sociological, involving (2.1.1) hypnosis, (2.1.2) mass hysteria, (2.1.3) memetic virus, (2.1.4) conditioning / disinformation, and (2.1.5) hoax; (2.2) psychological, including (2.2.1) false belief / misinterpretation, (2.2.2) fallible / screen memory, (2.2.3) psychosomatic disorder, (2.2.4) psychosis / hallucination, and (2.2.5) collective unconscious; and (2.3) neurological, including (2.3.1) psychedelics / pharmacology, (2.3.2) pathology / trauma, (2.3.3) neuro-divergence, (2.3.4) high intuition / functioning, and (2.3.5) "alien" implants.The third and final overarching category is metaphysical, involving three subcategories: (3.1) paranormal, including (3.1.1) psychotronics / extraordinary potential, (3.1.2) NDE / OBE, (3.1.3) disembodied spirit / consciousness, (3.1.4) incorporeal / bosonic lifeforms, (3.1.5) mediumship / possession, (3.1.6) new physics; (3.2) occult, including (3.2.1) hidden powers of nature, (3.2.2) ascended "masters," (3.2.3) "maya" / sensory delusion, (3.2.4) artificial reality / matrix simulation, (3.2.5) altered / alternate reality, and

(3.2.6) absolute world of archetypes; and (3.3) transcendental, including (3.3.1) “Gaia” / nature spirits, (3.3.2) angelic beings, (3.3.3) “djinn” / neutral powers, (3.3.4) demonic forces, (3.3.5) hierarchy of being, and (3.3.6) emanations of godhead.